



Fall 2025

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Project Title: Automated Ink Refill Machine
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Final Project Report
Mechanical Synthesis Course

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1 Introduction

The Automated Ink Refill Machine is designed to address the need for efficiency and sustainability in office and educational environments. Traditionally, refilling ink cartridges or markers is a manual, prone-to-error process. This project aims to design a mechanism that automates the plunging action required for precise ink delivery.

The primary objectives include:

- Developing a mechanism capable of a 1 minutes full cycle (refill and return).
- Ensuring high reliability and durability for prolonged use.
- Utilizing cost-effective materials like PLA and aluminum fasteners for fabrication.

2 Mechanism Design

The machine utilizes two lead screw mechanisms for the X and Z axes. This design was chosen for its inherent efficiency in converting rotational motion from a DC motor into the linear motion required for movement within the machine.

2.1 Key Components

The assembly consists of the following essential parts:

2.1.1 X-Axis

- **Linear Slider:** Responsible for guiding the motion of the markers along the X-axis. This component is mounted on a rail and moves in a linear fashion when activated by the lead screw.
- **Marker Holder:** A fixture that holds the markers in place during the refilling process. It is positioned and moved along the X-axis by the linear slider. This also act as the fulcrum (Strong point) for the process of removing the marker's tip.

2.1.2 Z-Axis

- **Marker's Tip Remover and Ejection:** A friction-used mechanism designed to automatically remove the tip of the marker. Once the marker is in place, the Z axis lower and this part grips on to the tips, utilizing friction to remove the tip. The A system of long shaft push the pen tip out of the remover, ensures the tip is ejected for refilling and then collected into a container bellow.
- **Ink Pump Needle:** Responsible for injecting the ink into the marker. This needle is connected to the ink pump assembly and is moved along the Z-axis to insert the ink into the marker after the tip has been ejected.

2.1.3 Electronics and Others

- **Ink Pump:** A critical component that pumps ink from the container into the marker. The pump is activated for a set duration to ensure the marker is adequately filled.
- **Machine Cover:** The outer shell that houses the components, protecting them and ensuring safe operation during the refilling process.

- ### 2.1.4 Detailed Component X Axis

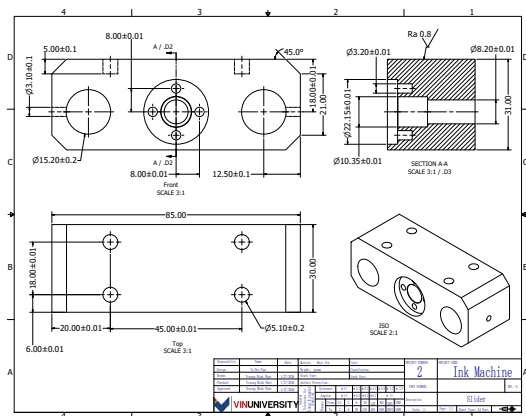


Figure 1: Base Plate

Figure 2: Slider

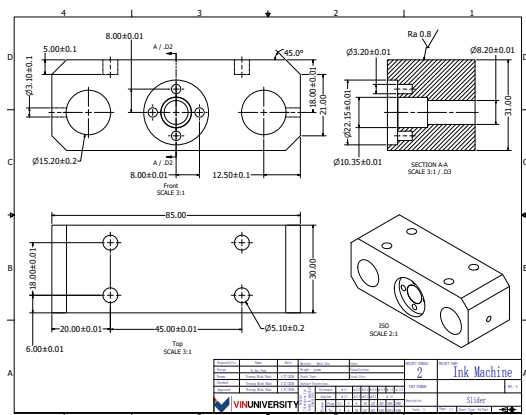


Figure 3: Motor Holder

Figure 4: Slider Plate

Figure 5: Overview of Slider in X axis



Technical drawing of a mechanical part, likely a bracket or plate, showing front and side views with dimensions and tolerances.

Front View (Top):

- Overall width: 164.10
- Overall height: 40.00
- Distance between centers of first and last holes: 127.00 ± 0.01
- Distance between centers of second and second-to-last holes: 39.98 ± 0.01
- Distance from right edge to center of last hole: 28.00 ± 0.1
- Distance from right edge to center of second-to-last hole: 28.00 ± 0.1
- Distance from right edge to center of last hole: 1.90 ± 0.01
- Radius of fillet: R20
- Scale: 1:1

Side View (Bottom):

- Overall width: 164.10
- Overall height: 40.00
- Distance between centers of first and last holes: 127.00 ± 0.01
- Distance between centers of second and second-to-last holes: 39.98 ± 0.01
- Distance from right edge to center of last hole: 28.00 ± 0.1
- Distance from right edge to center of second-to-last hole: 28.00 ± 0.1
- Distance from right edge to center of last hole: 1.90 ± 0.01
- Radius of fillet: R20
- Scale: 1:1

Isometric View (Right):

- Overall width: 164.10
- Overall height: 40.00
- Distance between centers of first and last holes: 127.00 ± 0.01
- Distance between centers of second and second-to-last holes: 39.98 ± 0.01
- Distance from right edge to center of last hole: 28.00 ± 0.1
- Distance from right edge to center of second-to-last hole: 28.00 ± 0.1
- Distance from right edge to center of last hole: 1.90 ± 0.01
- Radius of fillet: R20
- Scale: 1:1

Technical drawing of a mechanical part, showing three views: Front, Bottom, and Side. The part is a rectangular block with four circular holes. Dimensions are given in millimeters with tolerances.

Front View (ISO SCALE 2:1):

- Overall width: 12.00
- Overall height: 3.00
- Four circular holes with diameters: $\varnothing 5.00 \pm 0.2$, $\varnothing 8.50 \pm 0.2$, $\varnothing 7.10 \pm 0.2$, and $\varnothing 8.10 \pm 0.2$.
- Horizontal spacing between hole centers: 36.00 ± 0.2 , 40.00, 60.00 ± 0.2 , and 73.00.
- Vertical spacing: 28.00, 15.00, and 8.00.

Bottom View (ISO SCALE 2:1):

- Overall length: 73.00
- Overall width: 12.00
- Four circular holes with diameters: $\varnothing 5.00 \pm 0.2$, $\varnothing 8.50 \pm 0.2$, $\varnothing 7.10 \pm 0.2$, and $\varnothing 8.10 \pm 0.2$.
- Horizontal spacing between hole centers: 36.00 ± 0.2 , 40.00, 60.00, and 73.00.
- Vertical spacing: 28.00, 15.00, and 8.00.

Side View (ISO SCALE 2:1):

- Overall height: 30.50 ± 0.01
- Overall width: 15.00
- One circular hole with diameter: $\varnothing 3.10 \pm 0.2$.

ISO SCALE 2:1

Figure 9: Overview of Marker Holder Part

Technical drawing of a rectangular box with a lid, showing front, top, and side views with dimensions and a section view.

Front View: Shows the box with a lid. Dimensions: Total width 120.00, Total height 74.97, Lid height 231.93, Box height 240.93. Scale: 1:1.

Top View: Shows the box with a lid. Dimensions: Total length 231.93, Total width 240.93. Scale: 1:1.

Section A-A: Shows the lid and box assembly. Dimensions: Lid thickness 4.30 ± 0.01, Lid width 70.00, Lid height 54.00 M4, Lid angle 107.0°, Box height 25.00 ± 0.01, Box width 45.50 ± 0.01. Scale: 1:1 / 0.3.

ISO Scale 1:1: Shows the box with a lid. Dimensions: Total width 120.00, Total height 74.97, Lid height 231.93, Box height 240.93. Scale: 1:1.

Table:

Item	Quantity	Unit	Material	Notes
1	1	mm	Aluminum	Box body
2	1	mm	Aluminum	Lid
3	1	mm	Aluminum	Box body
4	1	mm	Aluminum	Lid
5	1	mm	Aluminum	Box body
6	1	mm	Aluminum	Lid
7	1	mm	Aluminum	Box body
8	1	mm	Aluminum	Lid
9	1	mm	Aluminum	Box body
10	1	mm	Aluminum	Lid
11	1	mm	Aluminum	Box body
12	1	mm	Aluminum	Lid
13	1	mm	Aluminum	Box body
14	1	mm	Aluminum	Lid
15	1	mm	Aluminum	Box body
16	1	mm	Aluminum	Lid
17	1	mm	Aluminum	Box body
18	1	mm	Aluminum	Lid
19	1	mm	Aluminum	Box body
20	1	mm	Aluminum	Lid
21	1	mm	Aluminum	Box body
22	1	mm	Aluminum	Lid
23	1	mm	Aluminum	Box body
24	1	mm	Aluminum	Lid
25	1	mm	Aluminum	Box body
26	1	mm	Aluminum	Lid
27	1	mm	Aluminum	Box body
28	1	mm	Aluminum	Lid
29	1	mm	Aluminum	Box body
30	1	mm	Aluminum	Lid
31	1	mm	Aluminum	Box body
32	1	mm	Aluminum	Lid
33	1	mm	Aluminum	Box body
34	1	mm	Aluminum	Lid
35	1	mm	Aluminum	Box body
36	1	mm	Aluminum	Lid
37	1	mm	Aluminum	Box body
38	1	mm	Aluminum	Lid
39	1	mm	Aluminum	Box body
40	1	mm	Aluminum	Lid
41	1	mm	Aluminum	Box body
42	1	mm	Aluminum	Lid
43	1	mm	Aluminum	Box body
44	1	mm	Aluminum	Lid
45	1	mm	Aluminum	Box body
46	1	mm	Aluminum	Lid
47	1	mm	Aluminum	Box body
48	1	mm	Aluminum	Lid
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67	1	mm	Aluminum	Box body
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74	1	mm	Aluminum	Lid
75	1	mm	Aluminum	Box body
76	1	mm	Aluminum	Lid
77	1	mm	Aluminum	Box body
78	1	mm	Aluminum	Lid
79	1	mm	Aluminum	Box body
80	1	mm	Aluminum	Lid
81	1	mm	Aluminum	Box body
82	1	mm	Aluminum	Lid
83	1	mm	Aluminum	

Technical drawing of a mechanical part, showing three views: Front View, Top View, and Isometric View.

Front View: Shows a rectangular part with a central slot. Dimensions include R4.00, 40.00 M4, 172.00 ± 0.01 , 20.00 ± 0.01 , 70.00, 58.00 ± 0.01 , 25.00 ± 0.01 , 5.00 ± 0.01 , 8.00 ± 0.01 , 13.50, and 20.00 ± 0.01 . Surface finish is Rz 20.

Top View: Shows the top of the part. Dimensions include 190.50, 26.1°, 13.00, 10.05, 20.50 ± 0.01 , and Ø4.00 M4. Surface finish is Rz 20. Section A-A is indicated.

Isometric View: Shows the 3D shape of the part. Dimensions include 20.00 ± 0.01 and 13.00. Surface finish is Rz 20.

Table:

Item	Quantity	Unit	Material	Notes
1	2	mm	Aluminum	Part 1
2	2	mm	Aluminum	Part 2
3	2	mm	Aluminum	Part 3
4	2	mm	Aluminum	Part 4
5	2	mm	Aluminum	Part 5
6	2	mm	Aluminum	Part 6
7	2	mm	Aluminum	Part 7
8	2	mm	Aluminum	Part 8
9	2	mm	Aluminum	Part 9
10	2	mm	Aluminum	Part 10
11	2	mm	Aluminum	Part 11
12	2	mm	Aluminum	Part 12
13	2	mm	Aluminum	Part 13
14	2	mm	Aluminum	Part 14
15	2	mm	Aluminum	Part 15
16	2	mm	Aluminum	Part 16
17	2	mm	Aluminum	Part 17
18	2	mm	Aluminum	Part 18
19	2	mm	Aluminum	Part 19
20	2	mm	Aluminum	Part 20
21	2	mm	Aluminum	Part 21
22	2	mm	Aluminum	Part 22
23	2	mm	Aluminum	Part 23
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26	2	mm	Aluminum	Part 26
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29	2	mm	Aluminum	Part 29
30	2	mm	Aluminum	Part 30
31	2	mm	Aluminum	Part 31
32	2	mm	Aluminum	Part 32
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37	2	mm	Aluminum	Part 37
38	2	mm	Aluminum	Part 38
39	2	mm	Aluminum	Part 39
40	2	mm	Aluminum	Part 40
41	2	mm	Aluminum	Part 41
42	2	mm	Aluminum	Part 42
43	2	mm	Aluminum	Part 43
44	2	mm	Aluminum	Part 44
45	2	mm	Aluminum	Part 45
46	2	mm	Aluminum	Part 46
47	2	mm	Aluminum	Part 47
48	2	mm	Aluminum	Part 48
49	2	mm	Aluminum	Part 49
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52	2	mm	Aluminum	Part 52
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54	2	mm	Aluminum	Part 54
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56	2	mm	Aluminum	Part 56
57	2	mm	Aluminum	Part 57
58	2	mm	Aluminum	Part 58
59	2	mm	Aluminum	Part 59
60	2	mm	Aluminum	Part 60
61	2	mm	Aluminum	Part 61
62	2	mm	Aluminum	Part 62
63	2	mm	Aluminum	Part 63
64	2	mm	Aluminum	Part 64
65	2	mm	Aluminum	Part 65
66	2	mm	Aluminum	Part 66
67	2	mm	Aluminum	Part 67
68	2	mm	Aluminum	Part 68
69	2	mm	Aluminum	Part 69
70	2	mm	Aluminum	Part 70
71	2	mm	Aluminum	Part 71
72	2	mm	Aluminum	Part 72
73	2	mm	Aluminum	Part 73
74	2	mm	Aluminum	Part 74
75	2	mm	Aluminum	Part 75

Figure 13: Pen Tip Collector Funnel

The technical drawing consists of two main views: a front view and an isometric view.

Front View: A long, thin mechanical part with a total length of 161.33. It features a central section with a width of 10.00 and a height of 1.00. The drawing includes dimensions for the overall length (161.33), the central section width (10.00), and the central section height (1.00). The drawing is labeled "Front SCALE 1:1".

Isometric View: A 3D representation of the part, showing its overall shape and dimensions. It includes dimensions for the overall length (161.33), the central section width (10.00), and the central section height (1.00). The drawing is labeled "ISO SCALE 1:1".

Title Block: A table containing the following information:

Part Name	Part No.	Rev.	Drawn By	Checked By	Approved By	Scale	Sheet No.	Sheet Total
						1:1	2	2
						Ink Machine Scale: 1:1 Date: 20/08/2023 Drawn By: A. B. C.		

Technical drawing of a mechanical part, showing front, top, and side views with dimensions and tolerances.

Front View: Shows a cross-section of the part. Dimensions include a total width of 30.00 mm, a central hole diameter of $\phi 12.15 \pm 0.2$ mm, and a hole depth of 27.50 mm. The part has a total height of 32.00 mm. The bottom flange has a thickness of 8.00 mm and a diameter of $\phi 22.15 \pm 0.2$ mm. The top flange has a thickness of 7.00 mm and a diameter of $\phi 15.20 \pm 0.2$ mm. The part is symmetrical about a vertical centerline.

Top View: Shows the top of the part. Dimensions include a total width of 85.00 mm, a central hole diameter of $\phi 15.20 \pm 0.2$ mm, and a hole depth of 45.00 mm. The part has a total length of 25.00 mm. The bottom flange has a thickness of 10.00 mm and a diameter of $\phi 22.15 \pm 0.2$ mm. The top flange has a thickness of 5.00 mm and a diameter of $\phi 15.20 \pm 0.2$ mm. The part is symmetrical about a horizontal centerline.

Side View: Shows the side of the part. Dimensions include a total height of 32.00 mm, a top flange thickness of 7.00 mm, and a bottom flange thickness of 8.00 mm. The part has a total width of 30.00 mm. The top flange has a diameter of $\phi 15.20 \pm 0.2$ mm. The bottom flange has a diameter of $\phi 22.15 \pm 0.2$ mm. The part is symmetrical about a vertical centerline.

Section A-A: A cross-section view of the part, showing a 105.6° angle and a surface finish of Ra 1.6. The section is labeled "SECTION A-A SCALE 1:1 / C3".

Isometric View: A 3D perspective view of the part, showing its overall shape and dimensions. The isometric view is labeled "ISO SCALE 1:1".

Table: A table with 10 columns and 2 rows. The first row contains the following data: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10. The second row contains the following data: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.

Footer: The footer contains the Vin University logo and the text "VIN UNIVERSITY".

Technical drawing of a mechanical part, likely a bracket or support, showing front, side, and detail views with dimensions and tolerances.

Front View: Shows a horizontal base with a central slot and two side slots. Dimensions include overall width 301.00, base width 270.00, and slot widths 67.35±0.1, 27.00±0.01, and 54.00±0.1. A fillet radius of R4.00±0.1 is indicated. A circular feature on the right has a diameter of 16.00 and a radius of R5.00. A detail callout 'Detail A' points to a specific feature.

Side View: Shows the profile of the part with a height of 16.00 and a radius of R5.00. A detail callout 'Detail B' points to a specific feature.

Detail A: A close-up view of the central slot area, showing dimensions 135.00, 11.49, 15.17, 11.49, 143.34, 140.31, 144.51, 90.00, 123.74, 10.05, 4.67, 88.10, 50.00, 15.50, and a fillet radius of R1.00. A detail callout 'Detail B' points to a specific feature.

Detail B: A close-up view of the circular feature, showing a diameter of 16.00 and a radius of R5.00.

Scale: The main views are at a scale of 1:1. The detail views are at a scale of 1:1.

Material: The part is made of **Aluminum**.

Finish: The part is finished with **Anodizing**.

Notes: The drawing includes a table of dimensions and tolerances, and a table of material and finish specifications.

Table 1: Dimensions and Tolerances

Dimension	Tolerance
301.00	±0.05
270.00	±0.05
67.35	±0.1
27.00	±0.01
54.00	±0.1
4.00	±0.1
16.00	±0.05
11.49	±0.05
15.17	±0.05
11.49	±0.05
143.34	±0.05
140.31	±0.05
144.51	±0.05
90.00	±0.05
123.74	±0.05
10.05	±0.05
4.67	±0.05
88.10	±0.05
50.00	±0.05
15.50	±0.05

Table 2: Material and Finish Specifications

Material	Finish
Aluminum	Anodizing

Figure 20: Overview of Slider in Z axis



2.1.8 Detailed Component of Machine Cover

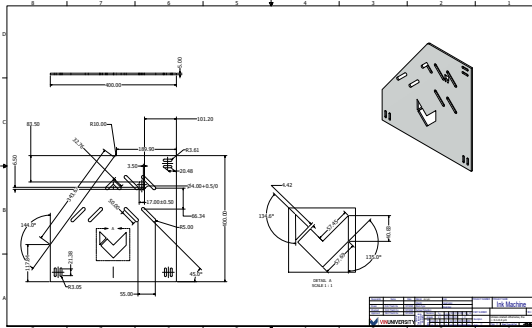


Figure 22: Side Plate

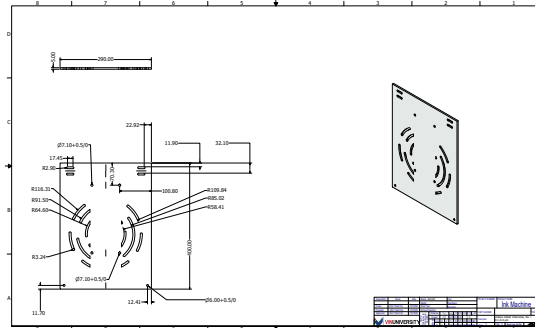


Figure 23: Back Plate

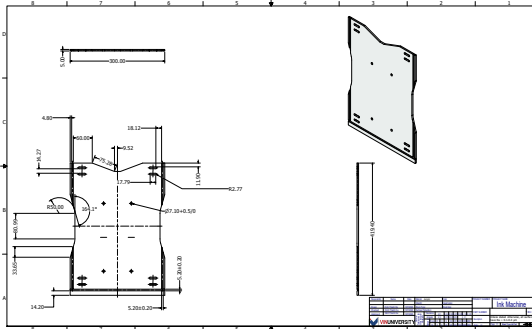


Figure 24: Base Plate

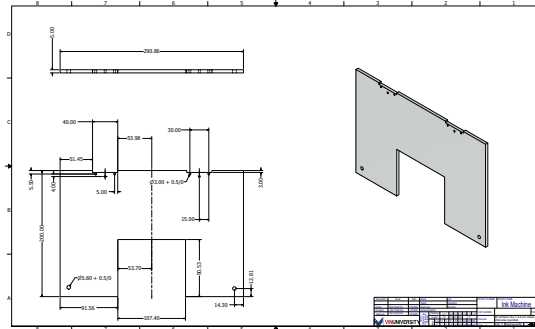


Figure 25: Roof Plate

Figure 26: Overview of Machine Cover

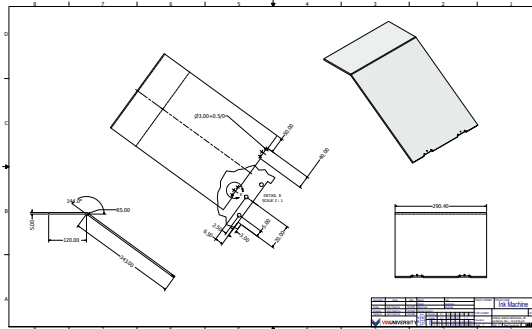


Figure 27: Door Plate

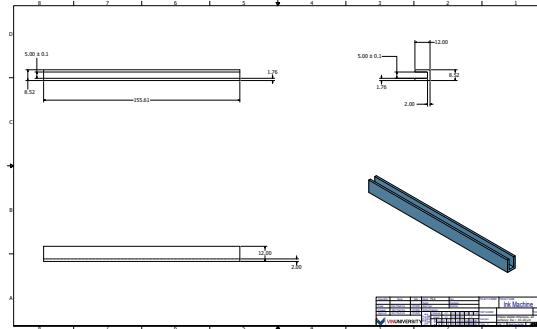


Figure 28: Shock Absorber (For the door)

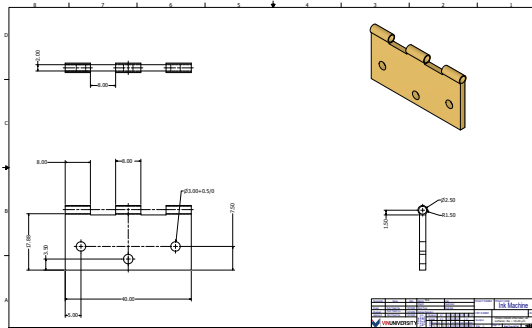


Figure 29: Hinge 1

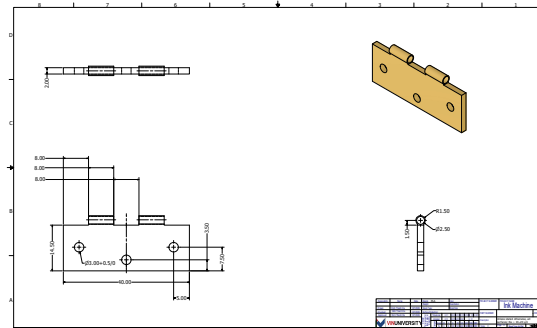


Figure 30: Hinge 2

Figure 31: Overview of Machine Door

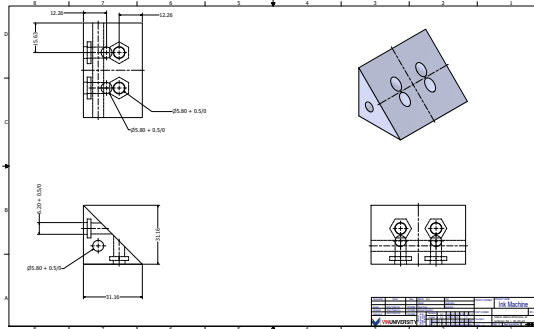


Figure 32: Square connector

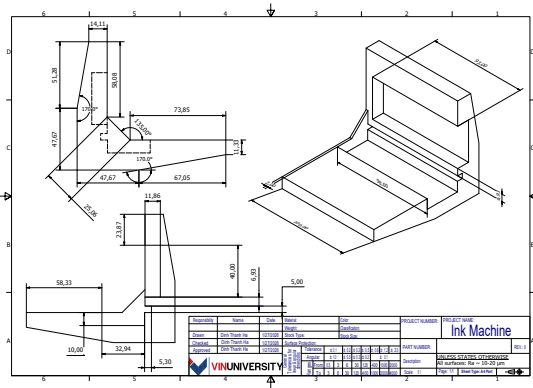


Figure 33: Machine Foot

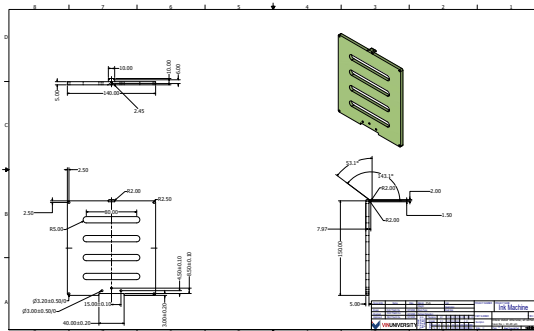


Figure 34: Electronic case cover

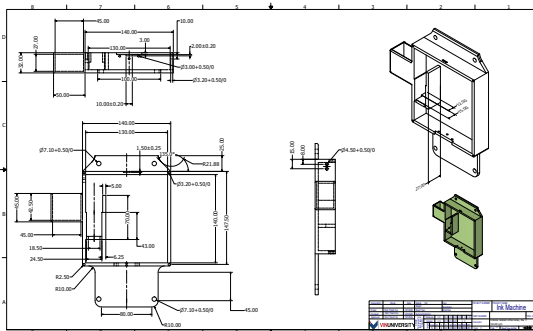


Figure 35: Electronic case

Figure 36: Other component

3 Physics Assessment and Calculations

3.1 Linear Displacement Calculation

This section converts a desired linear travel distance (in mm) into the corresponding number of stepper motor steps. The conversion depends on the lead screw pitch, the motor step resolution, and the microstepping setting.

3.1.1 General Step-to-Distance Relationship

Let:

- d = desired linear travel (mm)
- P = lead screw pitch (mm/rev)
- N_{motor} = full steps per motor revolution (steps/rev)
- M = microstepping factor (e.g., $M = 1$ full-step, $M = 2$ half-step, $M = 16$ for 1/16)

The number of commanded steps is:

$$N_{\text{steps}} = \frac{d}{P} \times N_{\text{motor}} \times M$$

Because the controller can only command an integer number of steps, the value is typically

rounded:

$$N_{\text{cmd}} \approx \text{round}(N_{\text{steps}})$$

3.1.2 X-Axis Calculations

For the X-axis:

$$P_x = 8 \text{ mm/rev}, \quad N_{\text{motor}} = 200 \text{ steps/rev}, \quad M = 1$$

So the axis resolution is:

$$\text{mm per step} = \frac{P_x}{N_{\text{motor}} \cdot M} = \frac{8}{200} = 0.04 \text{ mm/step}$$

or equivalently:

$$\text{steps per mm} = \frac{N_{\text{motor}} \cdot M}{P_x} = \frac{200}{8} = 25 \text{ steps/mm}$$

1) Start to Tip Removal ($d = 115 \text{ mm}$)

$$N_{\text{steps}} = 115 \times 25 = 2875 \text{ steps}$$

$$N_{\text{cmd}} = 2875 \text{ steps} \quad (\text{Controller setup: } 2820 \text{ steps})$$

The difference from the setup value is:

$$\Delta N = 2875 - 2820 = 55 \text{ steps} \Rightarrow \Delta d = 55 \times 0.04 = 2.2 \text{ mm}$$

This offset can be caused by practical factors such as: home switch offset, mechanical backlash, end-effector mounting offset, and the true effective pitch differing slightly from the nominal pitch.

2) Start to Tip Discharge ($d = 45 \text{ mm}$)

$$N_{\text{steps}} = 45 \times 25 = 1125 \text{ steps}$$

$$N_{\text{cmd}} = 1125 \text{ steps} \quad (\text{Controller setup: } 1120 \text{ steps})$$

Difference:

$$\Delta N = 5 \text{ steps} \Rightarrow \Delta d = 5 \times 0.04 = 0.2 \text{ mm}$$

3) Tip Ejection to Pump Station ($d = 12 \text{ mm}$)

$$N_{\text{steps}} = 12 \times 25 = 300 \text{ steps}$$

$$N_{\text{cmd}} = 300 \text{ steps}$$

Also, the shown setup value of 390 steps corresponds to:

$$d = 390 \times 0.04 = 15.6 \text{ mm}$$

This offset can be caused by practical factors similar as above.

3.1.3 Z-Axis Calculations

For the Z-axis:

$$P_z = 2 \text{ mm/rev}, \quad N_{\text{motor}} = 200 \text{ steps/rev}, \quad M = 1$$

Axis resolution:

$$\text{mm per step} = \frac{P_z}{N_{\text{motor}} \cdot M} = \frac{2}{200} = 0.01 \text{ mm/step}$$

$$\text{steps per mm} = \frac{N_{\text{motor}} \cdot M}{P_z} = \frac{200}{2} = 100 \text{ steps/mm}$$

1) Start to Tip Discharge ($d = 18.3 \text{ mm}$) **Note: there is an inconsistency in the original equation.** The distance is written as 18.3 mm but “20 mm” was used in the formula.

Using the stated distance $d = 18.3 \text{ mm}$:

$$N_{\text{steps}} = 18.3 \times 100 = 1830 \text{ steps}$$

$$N_{\text{cmd}} = 1830 \text{ steps} \quad (\text{Controller setup: } 2000 \text{ steps})$$

Difference:

$$\Delta N = 2000 - 1830 = 170 \text{ steps} \Rightarrow \Delta d = 170 \times 0.01 = 1.7 \text{ mm}$$

This may come from a deliberate safety clearance, a different real travel target, or additional offset due to tool length / mounting geometry.

2) Start to Tip Take and Pump ($d = 60 \text{ mm}$)

$$N_{\text{steps}} = 60 \times 100 = 6000 \text{ steps}$$

$$N_{\text{cmd}} = 6000 \text{ steps}$$

3.1.4 Practical Notes (Why Setup Values Differ)

In real systems, the final commanded steps are often tuned after measurement. Common reasons include:

- **Homing offset:** the home switch triggers before the true mechanical zero.
- **Backlash:** reversing direction introduces lost motion before load re-engages.
- **Compliance and load:** belts/couplers and mounts deform slightly under force.
- **Non-ideal pitch:** the effective lead screw pitch can deviate from the nominal value.
- **End-effector geometry:** the “functional point” (tip) is not exactly at the carriage reference.

For accuracy, measure the real travel per commanded steps and apply a calibrated steps/mm factor. We chose a 2 mm pitch on Z to get higher resolution and more holding/self-locking behavior, so the vertical axis moves smoothly in small increments and is less likely to back-drive under load. We chose an 8 mm pitch on X to get higher travel speed and efficiency for long horizontal moves, where gravity isn’t fighting the motion and fine resolution is less critical.

3.2 Tip Removal Force Requirement

The Engineering Challenge: In the previous version, the Z-axis frequently mis-stepped (Nema 17) during tip removal. The main reason was the large horizontal offset of the tip-removal head, which created a bending moment about the Z carriage. That moment increased side-loading on the guide system, increased friction, and therefore increased the torque demand on the lead screw beyond the motor limit.

3.2.1 Model and Assumptions (for a realistic engineering estimate)

- Nema 17 maximum holding torque: $T_{\max} = 0.36 \text{ N} \cdot \text{m}$
- Lead screw pitch (Z): $P = 2 \text{ mm/rev} = 0.002 \text{ m/rev}$
- Lead screw efficiency (typical for trapezoidal lead screw): $\eta \approx 0.30$
- Distance from screw axis to tool force line:

$$d_{\text{old}} = 123 \text{ mm} = 0.123 \text{ m}, \quad d_{\text{new}} = 73 \text{ mm} = 0.073 \text{ m}$$

- Two-guide/bearing spacing on the Z carriage (effective): $L \approx 30 \text{ mm} = 0.03 \text{ m}$
- Friction coefficient (misalignment + preload + dry sliding approximation): $\mu \approx 0.25$

3.2.2 Maximum Axial Force the Motor Can Provide

For a lead screw, the axial thrust from motor torque is approximately:

$$F_{\max} = \frac{2\pi\eta T_{\max}}{P}$$

Substituting:

$$F_{\max} = \frac{2\pi(0.30)(0.36)}{0.002} \approx 339 \text{ N}$$

So the Z-axis should ideally remain below about 339 N total equivalent load to avoid lost steps.

3.2.3 Physics of Offset: Moment - Side Load - Friction

During tip removal, the tool generates a vertical reaction force F . Because the force acts at a horizontal offset d , it produces a bending moment:

$$M = Fd$$

This moment must be reacted by the linear guides/bearings. A common approximation for the induced normal (side) load is:

$$N \approx \frac{M}{L} = \frac{Fd}{L}$$

This side load produces additional friction force on the Z motion:

$$F_{\text{fric}} = \mu N = \mu \frac{Fd}{L}$$

Therefore, the lead screw must overcome the **sum** of the removal force and the extra friction:

$$F_{\text{total}} = F + F_{\text{fric}} = F \left(1 + \mu \frac{d}{L} \right)$$

3.2.4 Old vs New Design: Total Force Demand

Old design ($d = 0.123$ m)

$$F_{\text{total, old}} = F \left(1 + \mu \frac{d_{\text{old}}}{L} \right) = F \left(1 + 0.25 \frac{0.123}{0.03} \right) = F(1 + 1.025) = 2.025F$$

New design ($d = 0.073$ m)

$$F_{\text{total, new}} = F \left(1 + \mu \frac{d_{\text{new}}}{L} \right) = F \left(1 + 0.25 \frac{0.073}{0.03} \right) = F(1 + 0.608) = 1.608F$$

This shows the redesign reduces the friction-amplification factor from 2.025 to 1.608:

$$\frac{F_{\text{total, new}}}{F_{\text{total, old}}} = \frac{1.608}{2.025} \approx 0.79$$

So the total Z load during removal is reduced by about 21% in this model, even though the bending moment alone scales by:

$$\frac{d_{\text{new}}}{d_{\text{old}}} = \frac{73}{123} = 0.59$$

3.2.5 Choosing a Plausible Tip Removal Force

The true tip removal force F is unknown. However, we know:

- **Old design mis-stepped** $\Rightarrow F_{\text{total, old}} > F_{\text{max}}$
- **New design worked reliably** $\Rightarrow F_{\text{total, new}} < F_{\text{max}}$

So we need:

$$2.025F > 339 \quad \text{and} \quad 1.608F < 339$$

Solving:

$$F > \frac{339}{2.025} \approx 167 \text{ N} \quad \text{and} \quad F < \frac{339}{1.608} \approx 211 \text{ N}$$

A reasonable conservative estimate that satisfies both is:

$$F \approx 190 \text{ N}$$

3.2.6 Check: Torque Demand Before vs After

Old design

$$F_{\text{total, old}} = 2.025(190) = 385 \text{ N} > 339 \text{ N}$$

Equivalent torque required:

$$T_{\text{req, old}} = \frac{F_{\text{total, old}} P}{2\pi\eta} = \frac{(385)(0.002)}{2\pi(0.30)} \approx 0.41 \text{ N} \cdot \text{m}$$

$$0.41 > 0.36 \text{ N} \cdot \text{m} \Rightarrow \text{Stall expected (matches observation)}$$

New design

$$F_{\text{total, new}} = 1.608(190) = 306 \text{ N} < 339 \text{ N}$$

$$T_{\text{req, new}} = \frac{(306)(0.002)}{2\pi(0.30)} \approx 0.32 \text{ N} \cdot \text{m}$$

$$0.32 < 0.36 \text{ N} \cdot \text{m} \Rightarrow \text{stable operation expected}$$

3.2.7 Conclusion

Reducing the tip-removal head offset from 123 mm to 73 mm reduces bending moment and, more importantly, reduces guide side-loading and friction. With a realistic lead screw efficiency and friction model, a plausible removal force around ~ 190 N explains why the old design exceeded motor capability (mis-stepping), while the new design stays within the Nema 17 torque limit and operates reliably.

4 Material Selection and Fabrication

Materials were selected based on cost-effectiveness and mechanical properties. The selected materials are chosen to balance performance, durability, and cost for the ink refiller machine.

- **PLA (3D Printed):**

- PLA (Polylactic Acid) is a biodegradable plastic commonly used in 3D printing due to its ease of use and cost-effectiveness. It is ideal for the non-load-bearing structural components of the machine such as the frame and covers. PLA is chosen for its relatively low cost, environmental benefits, and ease of fabrication.

- **CNC Mica Based:**

- CNC (Computer Numerical Control) machining is used to fabricate high-precision parts. Mica-based composites are used for parts requiring high thermal resistance and good electrical insulating properties. These materials are often selected for mechanical components that need high precision, such as parts of the pump assembly or the guide rail mechanisms.

- **Standard Metal Parts (Lead screw, Ball Bearing, Needle):**

- **Lead screw:** Typically made from stainless steel, the lead screw is crucial for the precise movement along the X and Z axes. Stainless steel provides the necessary strength, durability, and resistance to corrosion.
- **Ball Bearing:** Ball bearings are selected for their smooth operation and ability to handle axial and radial loads. They ensure that the movement of components like the linear slider is precise and efficient.
- **Needle:** The ink pump needle is typically made from stainless steel or another durable material, as it needs to be resistant to wear and corrosion while providing precision when filling the marker.

5 Timing and Cycle Analysis

5.1 Current Prototype Cycle Breakdown

The initial prototype demonstration established a baseline operational performance with a total cycle time of **34.4 seconds**. The cycle is divided into five distinct phases, primarily dominated by the Z-axis mechanical engagement and the ink infusion process. The following table provides a high-resolution breakdown of the recorded sequence of operations (S.O.P).

5.2 Detailed Phase Descriptions and Kinematic Logic

The following paragraphs detail the specific mechanical actions required to ensure successful automation and component safety:

Table 1: Prototype Sequence of Operations (S.O.P)

Phase	Mechanical Action Description	Duration (s)	Percentage (%)
t_1	Home Position to Tip Removal Alignment (X-axis)	3.6	10.5%
t_2	Tip Removal (Z-axis descent and extraction)	12.8	37.2%
t_3	Z-axis Discharge (Tip Ejection) and Station Transfer	5.2	15.1%
t_4	Ink Infusion Process (Primary Fluid Delivery)	11.8	34.3%
t_5	Machine Reset and Return to Initial Position	1.0	2.9%
Total	Cumulative Prototype Cycle Time	34.4 s	100%

Phase t_1 (Initial Alignment): This phase involves moving the marker holder from the home position to the alignment position for tip removal. During this interval, only the X-axis slider is active, while the Z-axis remains stationary to avoid interference with the housing or markers.

Phase t_2 (Tip Extraction): This is a Z-axis exclusive phase. The tip remover plate moves from its highest (home) position down to the engagement depth where it grips the marker tips. The axis then reverses to pull the plate and the old tips upward. At the end of this motion, the tips are physically secured against the pusher mechanism, completing the extraction.

Phase t_3 (Tip Ejection and Station Transfer): This phase coordinates both axes. First, the X-axis moves slightly outward to align the extracted tips with the collection tray. The Z-axis then descends (approximately 2000 steps) to push the old tips out of the grippers and into the tray. Finally, the X-axis continues moving to the ink refill station to prepare for the infusion phase.

Phase t_4 (Ink Infusion): Motion is again limited to the Z-axis, which descends so the needles are entirely inserted into the marker bodies. The ink pump is then activated for 10 seconds in the current demo version, though this is increased to 20 seconds in the adjusted high-precision version. After pumping, the Z-axis returns to its highest position.

Phase t_5 (Cycle Reset): The X-axis slider (marker holder) moves back to the home position at high speed to conclude the cycle and allow the user to retrieve the refilled markers.

5.3 Observations and Fluid Dynamics Limitations

During preliminary testing with high-viscosity industrial ink, a significant “spill and splatter” failure was identified during Phase t_4 . At the current prototype pump rate, the volumetric flow Q_{pump} exceeds the capillary absorption capacity Q_{cap} of the marker’s internal fiber reservoir. This imbalance results in several critical failure modes:

- **Ink Backflow:** Excess fluid causes contamination of the marker holder and internal components.
- **Air Entrapment:** Rapid pumping introduces air bubbles into the reservoir, reducing refill quality and affecting ink longevity.
- **System Stress:** High pressure on the pump tubing leads to premature mechanical fatigue and an increased risk of hardware leakage.

5.4 Optimization and Product Specifications

To transition from a prototype to a market-ready industrial product, the cycle time has been recalibrated by prioritizing precision delivery over raw speed. By reducing the pump rate, we

ensure consistent laminar flow during the refill process.

In the final product version, the infusion phase (t_4) is extended to **15.0–20.0 seconds**. This modification ensures a mess-free, professional-grade refill. The total cycle time for the commercial version is modeled as:

$$T_{total} = t_{alignment} + t_{extraction} + t_{transfer} + t_{optimized_pump} + t_{reset} \quad (1)$$

The target cycle time for the commercial version is estimated at **40.0–45.0 seconds**. While this increases the cycle duration by approximately 25%, it results in a **100% reduction** in ink waste and mechanical failure due to fluid overflow. This trade-off is consistent with industrial standards for high-precision automated fluid handling systems.

5.5 Observations and Limitations

During preliminary testing with high-viscosity industrial ink, a significant "spill and splatter" issue was identified during Phase t_4 . At the prototype pump rate, the volumetric flow exceeded the absorption capacity of the pen's internal reservoir. This resulted in:

- **Ink Backflow:** Excess fluid causing contamination of the pen holder.
- **Air Entrapment:** Rapid pumping introducing bubbles, reducing refill quality.
- **System Stress:** Higher pressure on the pump tubing leading to premature mechanical fatigue.

6 Control System

6.1 Control Diagram and Logic

The machine's cycle is governed by an electronic control system, typically using an Arduino or a PLC (Programmable Logic Controller), which coordinates the motion of the system and the refilling process. The control system ensures that the marker is accurately refilled in a reliable and automated manner. The following logic governs when a new cycle is initiated and how the system operates step by step:

- **Input Trigger:** The cycle begins when the system detects the presence of a marker. This can occur through two possible input methods:
 - **Proximity Sensor:** A proximity sensor detects the marker when it is placed on the shelf. This sensor ensures that only the markers that are properly positioned will trigger the cycle. The sensor may be positioned on the marker shelf or within the mechanism that receives the markers.
 - **Manual Button Press:** In case the user needs to manually trigger the cycle (e.g., for maintenance or specific needs), a button is used to activate the process. Once pressed, the system initiates the start of the cycle, bypassing any sensor input.
- **Cycle Start:** Once the input trigger is detected, the system begins the cycle. The following actions occur to control the movement of the machine:
 - **Activation of Stepper Motors:** The control system activates two stepper motors, one for the X-axis and one for the Z-axis, which move the markers along the designated paths. The motors are precisely controlled using pulse signals that dictate the exact number of steps each motor needs to take to move the markers to their correct

positions. These positions are determined by the machine's design and the specific task to be performed (e.g., aligning with the tip remover or ink pump).

- Movement Logic: The X-axis moves the marker into position in front of the tip remover. After the alignment, the Z-axis moves downward to perform the removal and other tasks like ink refilling.
- **Pump:** The ink pump is activated when both the X and Z axes are in their designated positions. This occurs under the following conditions:
 - Marker Positioning: The X and Z motors position the marker to ensure that the tip remover is properly aligned with the marker tip. Once this alignment is confirmed, the Z-axis lowers to remove the tip automatically using friction-based grip.
 - Ink Injection: Upon successful tip removal, the Z-axis moves towards the ink pump station, and the pump is activated to inject ink into the marker for 5 seconds. The system uses a flow control valve or a similar mechanism to regulate the amount of ink dispensed into the marker to avoid overfilling.
- **Cycle Completion:** After the ink is dispensed, the system checks whether the process is complete. The cycle completion is determined through the following steps:
 - Limit Switch Detection: A limit switch or a sensor at the end of the movement range ensures that the system has reached the final position for both axes. When the end of travel is reached, this switch or sensor will send a signal to the control system, indicating that the marker is in the correct position, or the ink has been injected.
 - Encoder Feedback: If encoders are used, the system monitors the motor position through encoder counts, ensuring that the motors have completed the required number of steps. Once the required displacement has been achieved, the system can reset and prepare for the next cycle.
 - Reset Mechanism: After detecting the end of the process, the system sends a signal to reset the system, bringing the marker holder back to its starting position or into a safe state. This can be accomplished using limit switches or feedback from the encoders that track the position of the system components.
- **Safety Features:**
 - Error Detection: The system constantly monitors for any errors, such as blocked movements or malfunctions in the sensors. If an error is detected, the cycle is paused, and an error signal is generated to alert the user.
 - Overload Protection: To avoid motor damage, the system includes safety features to stop the motors if an excessive load is detected during the motion. This prevents mechanical failures and ensures the longevity of the system.

This control system ensures that the refilling cycle is both automated and efficient, allowing for consistent performance. The integration of feedback loops, sensors, and actuators guarantees precision while minimizing user intervention.

6.2 Working Steps Illustration

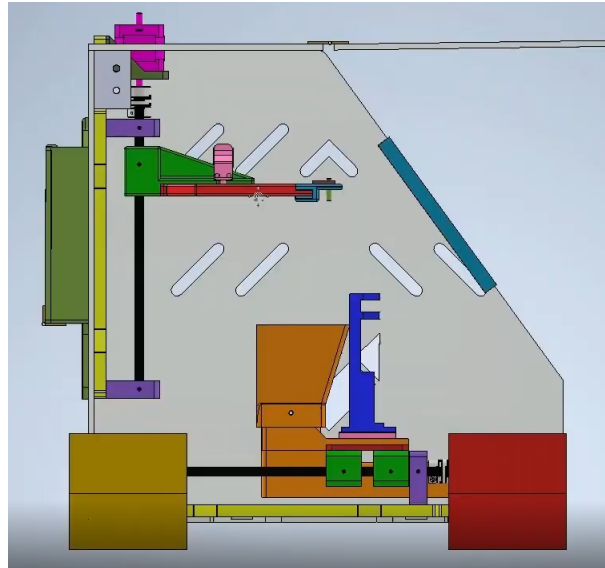


Figure 37: Initial State (User insert Markes)

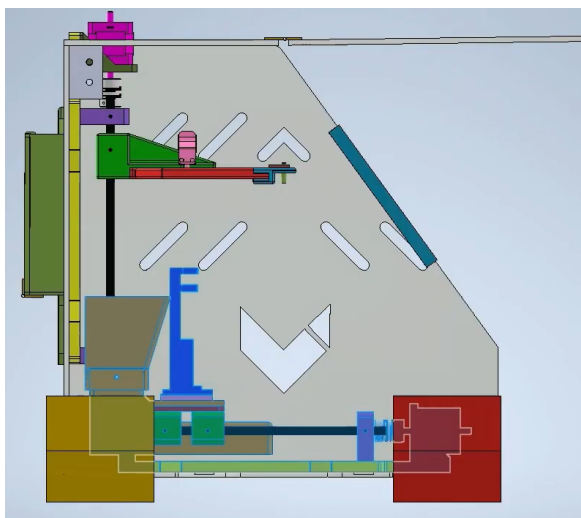


Figure 38: X Axis goes in to tip remover

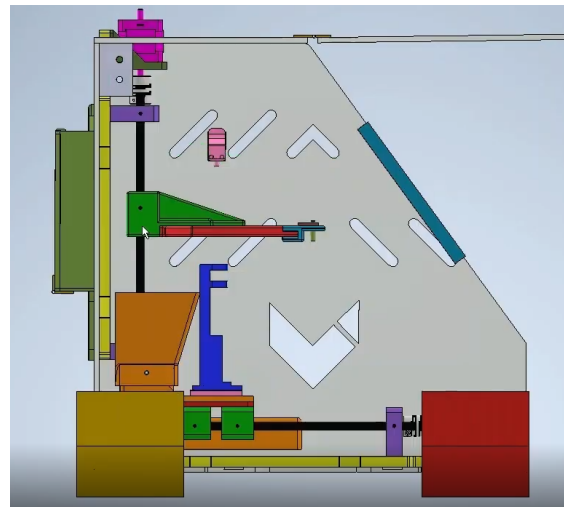


Figure 39: Z Axis goes down to grab the markers' tips

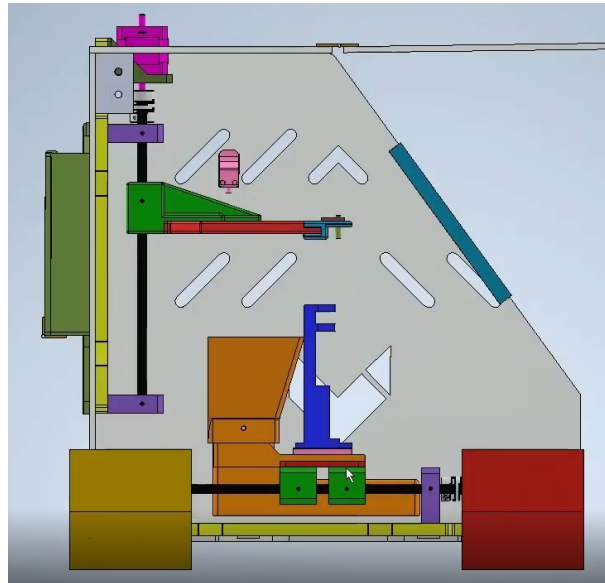


Figure 40: Marker

Figure 41: Overview of Marker Holder Part

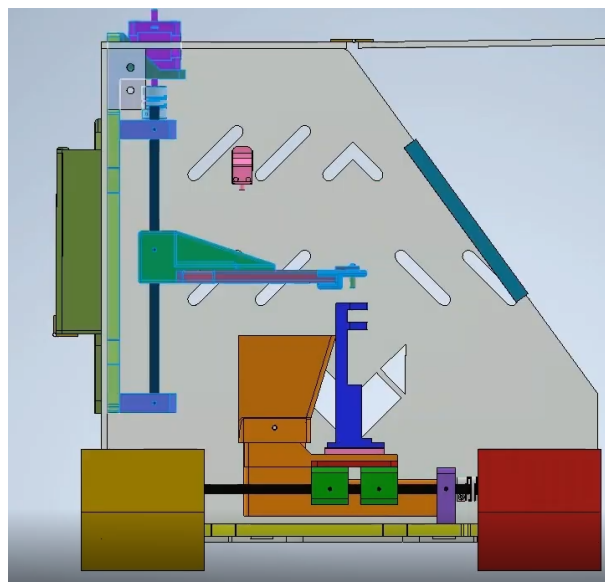


Figure 42: X axis goes out, Z axis goes down, Pump activated

6.3 Working Flow Chart

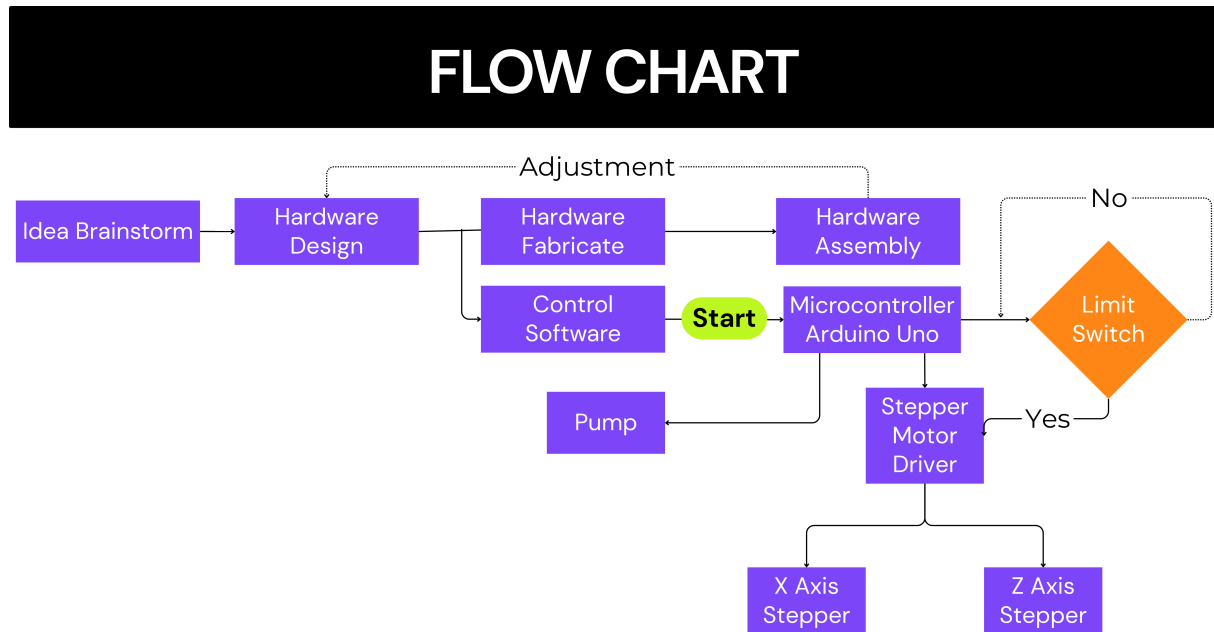


Figure 43: Machine Flow Chart

7 User Manual

7.1 Operating Instructions

To ensure the longevity of the machine and the safety of the user, the following procedures must be followed strictly:

1. **Pen Installation:** Insert the pens into the designated pen holders with caution and a gentle touch. Align the bottom of the pen with the corresponding hole in the holder. Gently apply downward force until a distinct "snap" is heard, indicating the pen is securely locked in the correct position.
2. **Safety Enclosure:** Close the machine door before initiation. This safety measure is mandatory to prevent injury from moving parts and to contain potential malfunctions, such as ink spills or component snapping, which could cause damage or mess.
3. **Cycle Initiation:** Press the "Start" button located on the exterior of the chassis.
 - *Note:* While the current prototype requires manual activation, the final product version features an integrated external button for improved ergonomics.
4. **Automated Process:** The machine will execute the full refill cycle automatically, moving from tip removal to precise ink injection.
5. **Pen Removal:** Once the cycle is complete and the mechanism has returned to its initial "home" position, hold the pen holder securely. Gently tilt the pen to remove the top portion first, then extract the pen entirely and install a new tip.

7.2 Maintenance Standards

The following maintenance schedule is based on industrial standards for precision mechanical synthesis to ensure maximum efficiency and prevent hardware failure:

Table 2: Planned Maintenance Schedule

Component	Action	Frequency
Slider Rails & Lead Screw	Lubrication (High-viscosity grease)	Every 100 cycles
Refill Needle	Inspection & Replacement	Weekly
Pump Tubing	Component Replacement	Every 100 cycles
Fasteners	Torque Check (Vibration check)	Monthly

- **Lubrication:** Apply industrial-grade lubricant to the slider rails and lead screw every 100 cycles to reduce friction coefficients and prevent motor strain.
- **Fluid Delivery Path:** Replace the refill needle weekly to prevent ink coagulation and "ink jams."
- **Peristaltic/Pump Efficiency:** The pump tube must be replaced every 100 cycles. Over time, tube elasticity decreases (mechanical fatigue), which can lead to inaccurate ink volume delivery.
- **Contamination Control:** Perform a "dry run" (cycle without ink) after every needle replacement to ensure the alignment of the slider mechanism.

7.3 Safety and Troubleshooting

- **Emergency Stop:** In the event of an abnormal noise or ink leakage, immediately disconnect the power supply.
- **Standardized Warning:** Do not attempt to bypass the door safety sensors or open the enclosure while the linear stage is in motion.

8 Engineering Challenge and Improvement

8.1 Tip Removal Force Requirement

The Engineering Challenge: In the previous version, the Z-axis mechanism experienced frequent mis-stepping in the Nema 17 motor. This occurred because the "vit me" (lead screw) had to overcome a high resistive torque caused by the horizontal offset of the tip-removal head.

Physics of the Improvement: The tip-removal process generates a vertical reaction force F as the tool engages the pen. Because this force is applied at a distance d from the lead screw axis, it creates a Bending Moment M :

$$M = F \times d$$

Old Design ($d = 123 \text{ mm}$): The large offset created a high moment, causing significant friction on the guide rails and high torque demand on the motor.

New Design ($d = 73 \text{ mm}$): By reducing the distance by 40%, we directly reduced the bending moment and the resulting frictional forces.

The torque requirement due to the offset was reduced to:

$$\frac{73}{123} = 0.59 \quad (59\%) \quad \text{of the original value}$$

This provides a 41% increase in the mechanical safety margin.

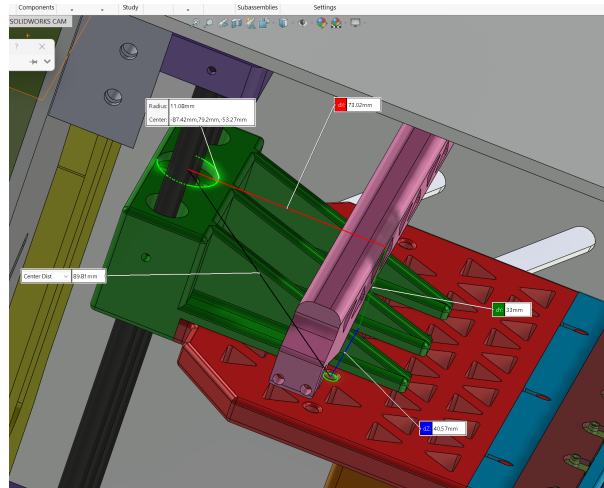


Figure 44: Fore and Moment Adjustment

8.2 Pump adjustment

The Engineering Challenge:

- **Placement:**
 - Where to put the pump so that it is neither too exposed to the exterior on the user's end, but not too closed off where it is hard to take it out, either for repairing or switching to another component(s)
 - The lack of adhesive usage and friction fitting, due to requirements, also contribute to the problem
 - And, hence, the design of the pump container took a few of completely different iterations to finally come to the final design
- **Electronics:**
 - **Components:** A lot of times, parts failed randomly and unexpectedly without any clear explanation
 - **Schematics:** It can be unintuitive to figure out how components interacting with each others

Improvement:

- **Full set:** Upgrade to a full set of 6 pumps, per the initially expected requirements
- **Weaker pump:** It is needed to weaken the flow of the ink. We could do that by shrinking the pipe diameter, for example.
- **Customizability, in terms of control:** Upgrade from simple plugging in and out of the power source to having a programmable and configurable Graphical User Interface

(GUI), or at least a simple and intuitive Terminal User Interface (TUI) (with full manual page). From either of these interfaces, users can control parameters of the machines, for examples: amount of ink being pump, which pen is currently slotted into places, color of inks, etc.

- **Watch out for clogs:** The ink of whiteboard markers is made from pigments, unlike the common permanent dye ink for fountain pens. Due to such nature, sometimes it has particles, if leave dried after a couple of days, might clog the either or both the pipeline or the pump itself. To prevent this, we can rinse the pumps with warm water for at least once per month or when clogging is suspected.

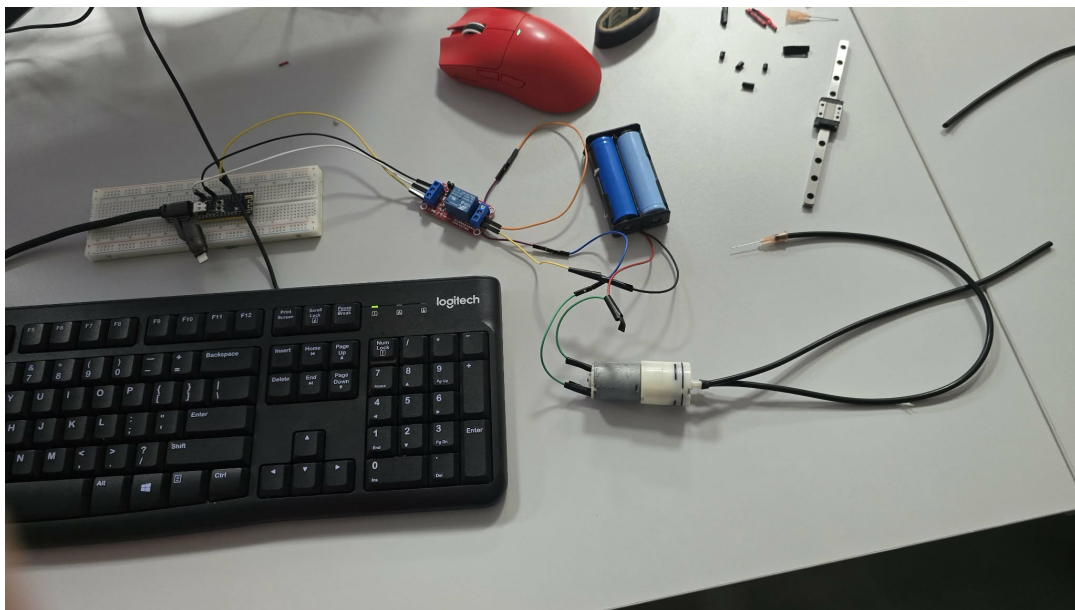


Figure 45: Pump

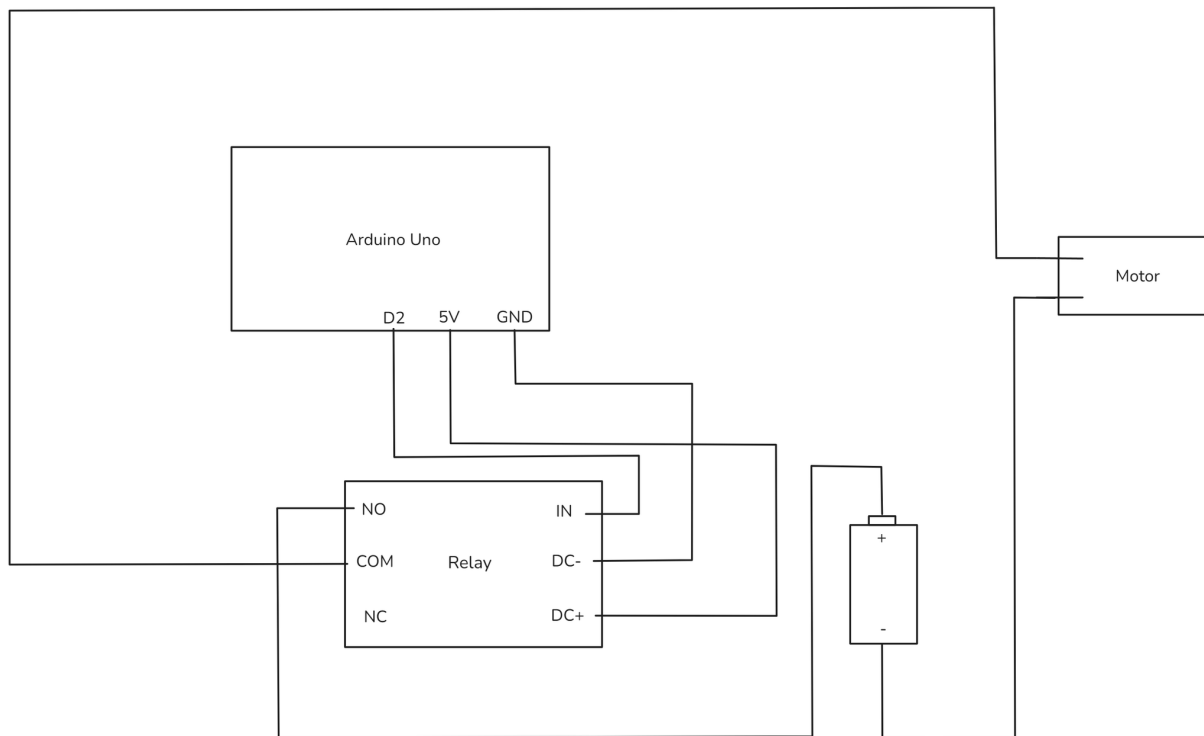


Figure 46: Pump Schematic

8.3 Electronic Problem

The Engineering Challenge: During the development of the automated marker ink refiller, several challenges were encountered in the electronic system's design and implementation. These problems arose primarily from the limitations of the prototype's wiring, current handling capabilities, and overall system reliability. The key issues were as follows:

- **Bad Breadboard:** The initial testing and prototyping used a breadboard for wiring connections, which introduced several issues. Breadboards are not designed for handling high current and can lead to unstable connections, resulting in unreliable performance. They also cause intermittent issues such as poor conductivity or loose connections, which can disrupt the operation of the motors and sensors.
- **Easy to Overcurrent Driver:** The motor drivers, particularly those controlling the stepper motors for the X and Z axes, were often subjected to overcurrent conditions. This happened because the system drew more current than the drivers could handle, leading to overheating and driver failure. In some cases, we had to replace the motor drivers multiple times due to damage caused by excess current.
- **Hot Wires Due to Large Current:** The wiring used for the motors and electronic components was not sufficiently rated for the current demands. This caused the wires to heat up during operation, which posed a risk of melting insulation, short circuits, or fire. The wires were getting excessively hot, especially when motors were running at higher currents during peak operation.
- **Routing Wires Was Difficult Due to Limited Space:** The compact design of the machine left limited space for routing the various wires that connected the motors, sensors, pump assembly, and electronics. This made it difficult to organize the wires efficiently, leading to tangled or messy wiring that added to the complexity of the system. Poorly

routed wires could also lead to accidental shorts or interference between signals.

Improvement: To overcome these challenges, we implemented several changes to improve the electronic system's reliability, safety, and performance. The improvements focused on better managing current, organizing wiring, and ensuring stable connections throughout the system.

- **Remove All Breadboards, Replace with Direct Soldering:** We eliminated the use of breadboards altogether, as they were unsuitable for handling the power and current required by the motors and other components. Instead, we switched to direct soldering of components, which provided more stable and secure electrical connections. This change significantly improved the reliability of the electrical system, reducing issues like loose connections or short circuits.
- **Adjust Current Based on Load:** We realized that different components required different current levels based on their load. For example, the Z-axis motor required more current compared to the X-axis motor, as it needed to lift and move heavier parts. To address this, we adjusted the current supply to each motor individually to match their specific requirements, thereby preventing overcurrent and overheating issues. This also improved the longevity of the motor drivers.
- **Divide Separate Wires for 2 Motors:** Initially, both motors shared the same set of wires, which caused issues with power distribution and current flow. We split the wiring for the two motors into separate circuits, each with its own dedicated wire and power source. This prevented power surges from one motor affecting the other and reduced the risk of overloading a single wire or driver. It also allowed for more precise current control for each motor.
- **Use Cable Clips to Organize the Wires:** To address the issue of limited space for routing wires, we used cable clips to neatly organize and secure the wiring. The clips ensured that wires were arranged in a tidy, efficient manner, minimizing the risk of accidental shorts or tangling. By properly organizing the wires, we also improved airflow and reduced the chance of overheating due to tangled or bundled wires. This made maintenance and troubleshooting much easier, as it was clear which wires went where, and reduced clutter inside the machine.
- **Use Higher Gauge Wires for High Current Components:** We upgraded the wiring for components requiring higher current (such as the motors and pump assembly) to a higher gauge wire, which can handle more power without heating up. This change ensured that the wires did not overheat and significantly improved the safety of the system.

These improvements allowed the machine to operate more reliably, efficiently, and safely. The transition from breadboard-based connections to soldered circuits, coupled with better current management and organized wiring, helped avoid overheating, overcurrent, and wiring issues. The machine's overall performance has been significantly enhanced, ensuring long-term durability and reduced risk of system failure.'

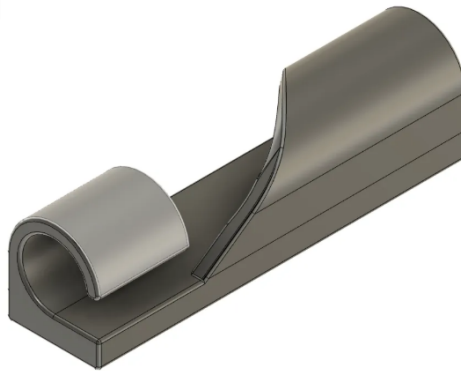


Figure 47: Cable Clip

8.4 Cover side plate shape

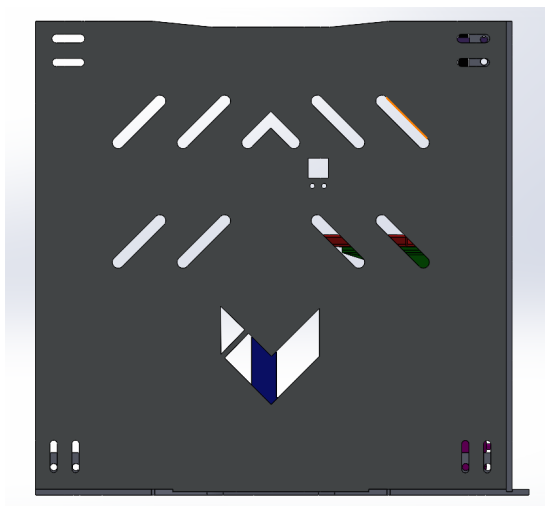


Figure 48: Before Modification

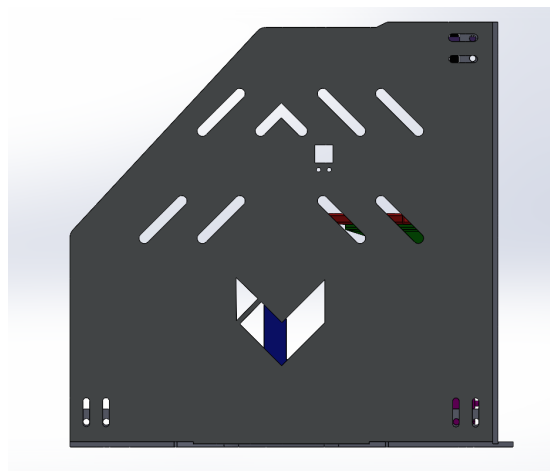


Figure 49: After Modification

Why the Change Was Necessary: The modification to the cover side plate was made to address structural limitations and improve functionality. The original design (Figure 26) had weak points that could lead to stress concentrations and hinder component clearance. The new shape enhances the structural integrity, optimizes space, and simplifies assembly and maintenance.

How the Change Affects the Machine Cover:

- **Aesthetic Improvement:** The new design provides a cleaner, more modern appearance with better symmetry and refined cutouts.
- **Improved Durability:** The adjusted shape removes sharp edges and improves the cover's strength, contributing to better safety and longevity.
- **Better Component Alignment:** The modification allows for better clearance of internal mechanisms and components, improving functionality and heat dissipation.
- **Future Flexibility:** The design facilitates easier integration of additional parts and simplifies cable management.

9 Static Structure Analysis

9.1 Introduction

This test is performed to test how many times can we insert the pen into one of the rack within the assembly, or in other words, its life cycles and other potentially helpful parameter.

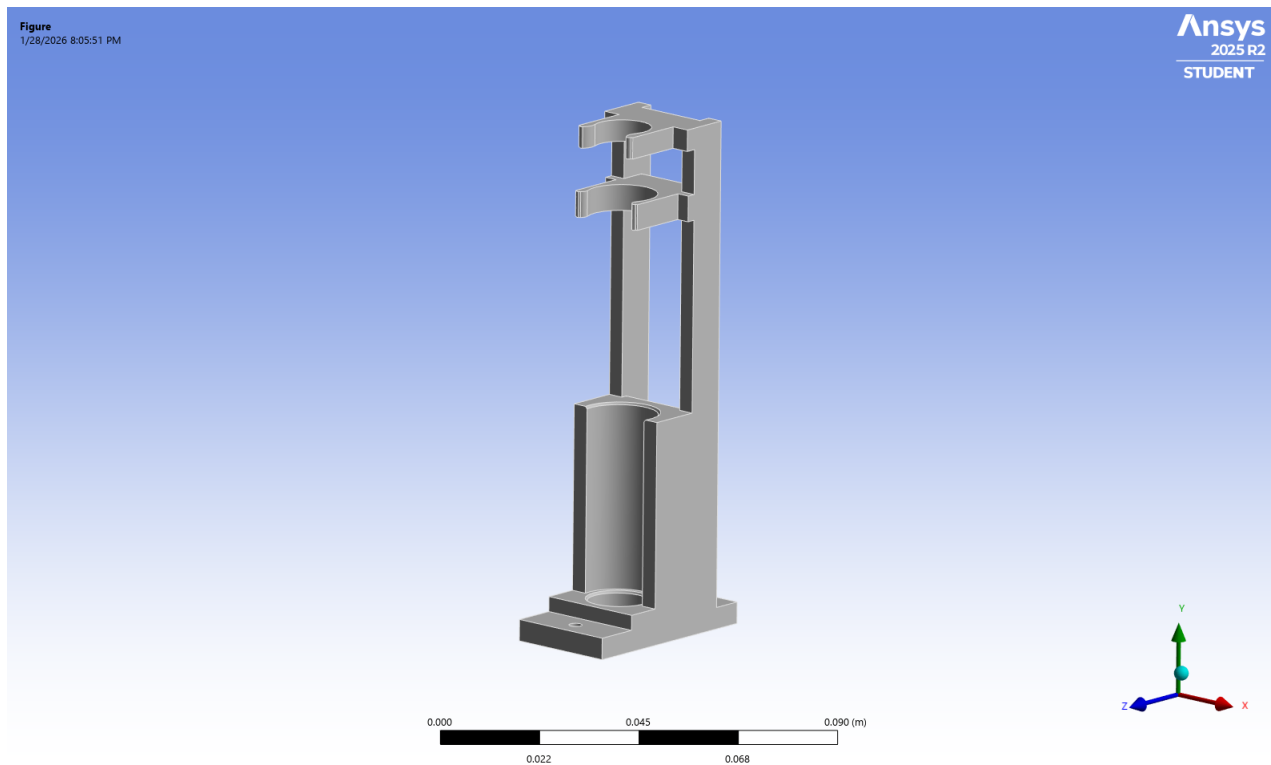


Figure 50: Model Overview in ANSYS

9.2 Material Properties

Property	Value
Material: PLA	
Density	1250 kg m^{-3}
Young's Modulus	$3.45 \times 10^9 \text{ Pa}$
Poisson's Ratio	0.39
Bulk Modulus	$5.2273 \times 10^9 \text{ Pa}$
Shear Modulus	$1.241 \times 10^9 \text{ Pa}$
Tensile Yield Strength	$5.41 \times 10^7 \text{ Pa}$
Tensile Ultimate Strength	$5.92 \times 10^7 \text{ Pa}$

Table 3: PLA Material Properties [1]

Alternating Stress (Pa)	Cycles
1.55×10^8	250
1.30×10^8	350
1.05×10^8	550
8.20×10^7	800
6.00×10^7	1200
3.70×10^7	1700

Table 4: PLA S-N Curve Data [2]

9.3 Setting up test environment

Meshing: Made using automatic mesh generation by ANSYS Mechanical

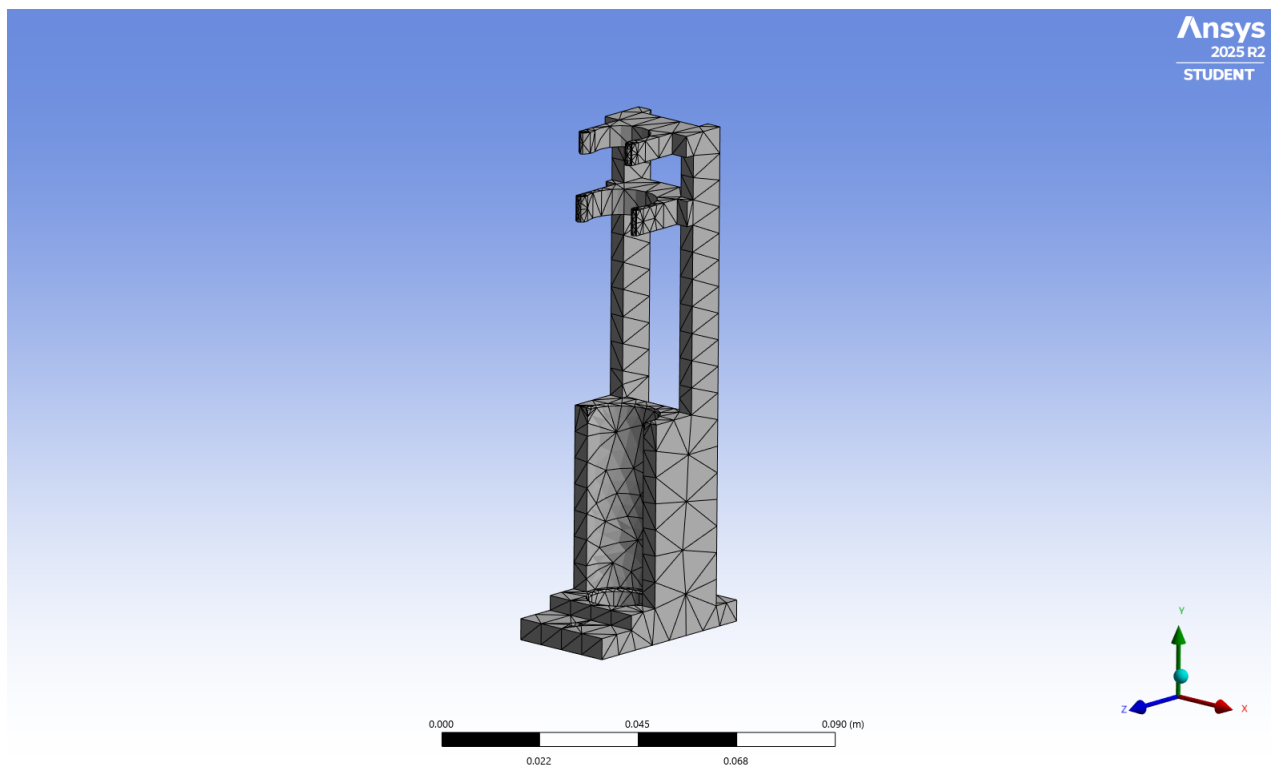


Figure 51: Meshing showcase

Forces are set up as below:

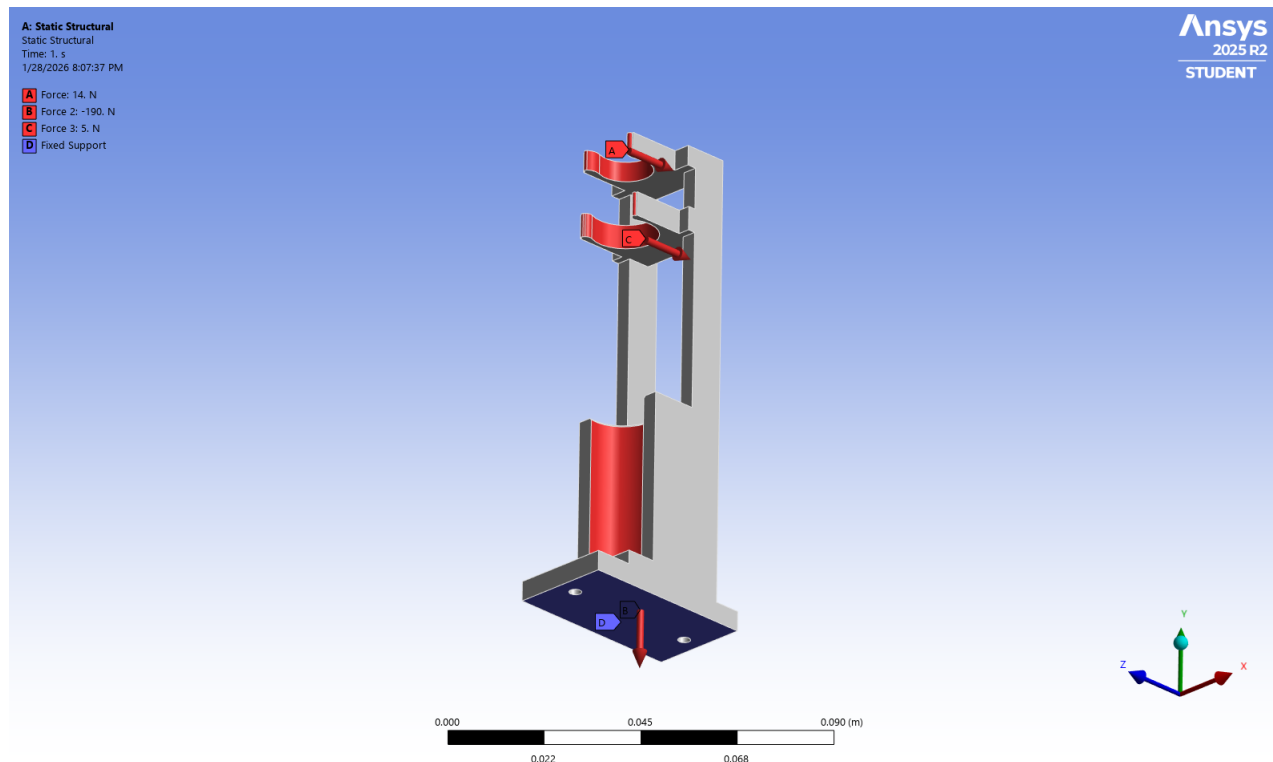


Figure 52: Force applied on the model and Fixed Support

In which:

- **Component A:** Force of 14N is applied to the tip of the upper clips, represent the forces made by the user to the insert the marker inside the rack [3], where it was assumed to be equivalent to the maximum compression force made by the thumb.
- **Component B:** Force of -190N is the weight of the pen applied on the surface of the slot.
- **Component C:** 5N force applied to the inside walls as the light-snapped force when the pen got inserted inside.
- **Component D:** Fixed support, which is the blue surface area underneath the model.

9.4 Total Deformation and Equivalent Stress

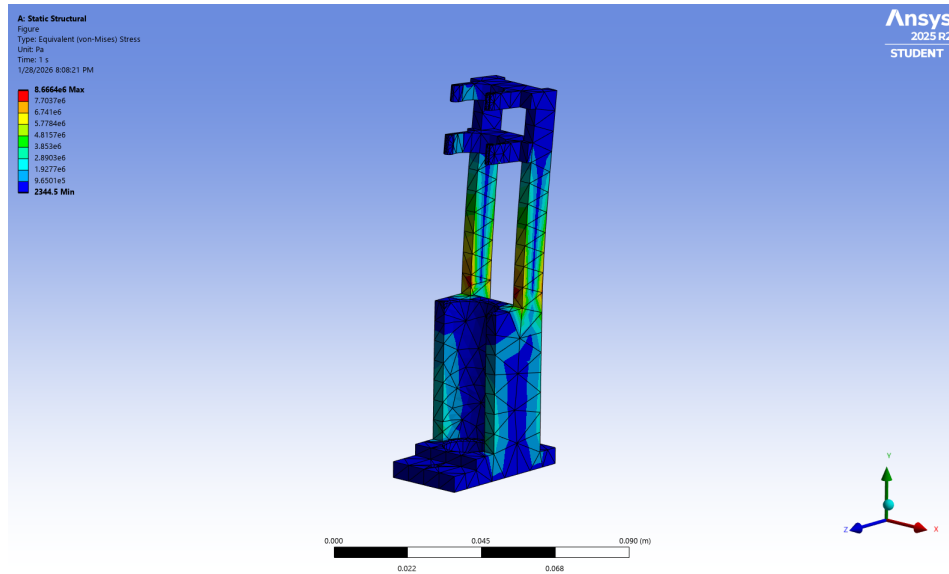


Figure 53: Equivalent stress on the model

Overall high stress around the middle beam and its edges where the user would have to likely bend it a little to push the pen inside the socket. While it does not exceed the yield strength of the material, however, the force concentration could be improved by filleting the 4 edges of the said beam.

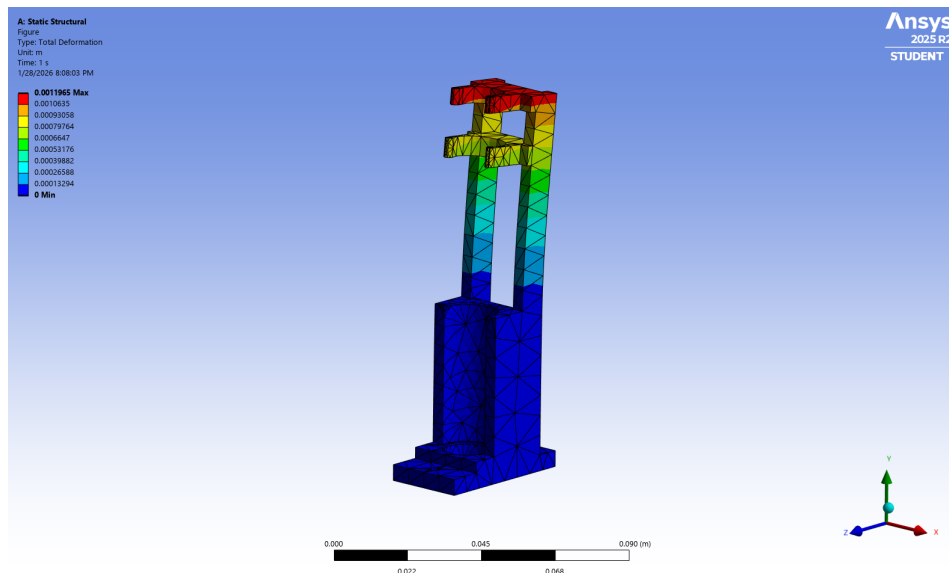


Figure 54: Total deformation of the model

The total deformation result is as expected, because this is also how the model actually bends in real life when trying to insert the marker inside. Due to the markers' dimension requirements itself, there aren't much that can be change in terms of socket's structure.

9.5 Fatigue Test

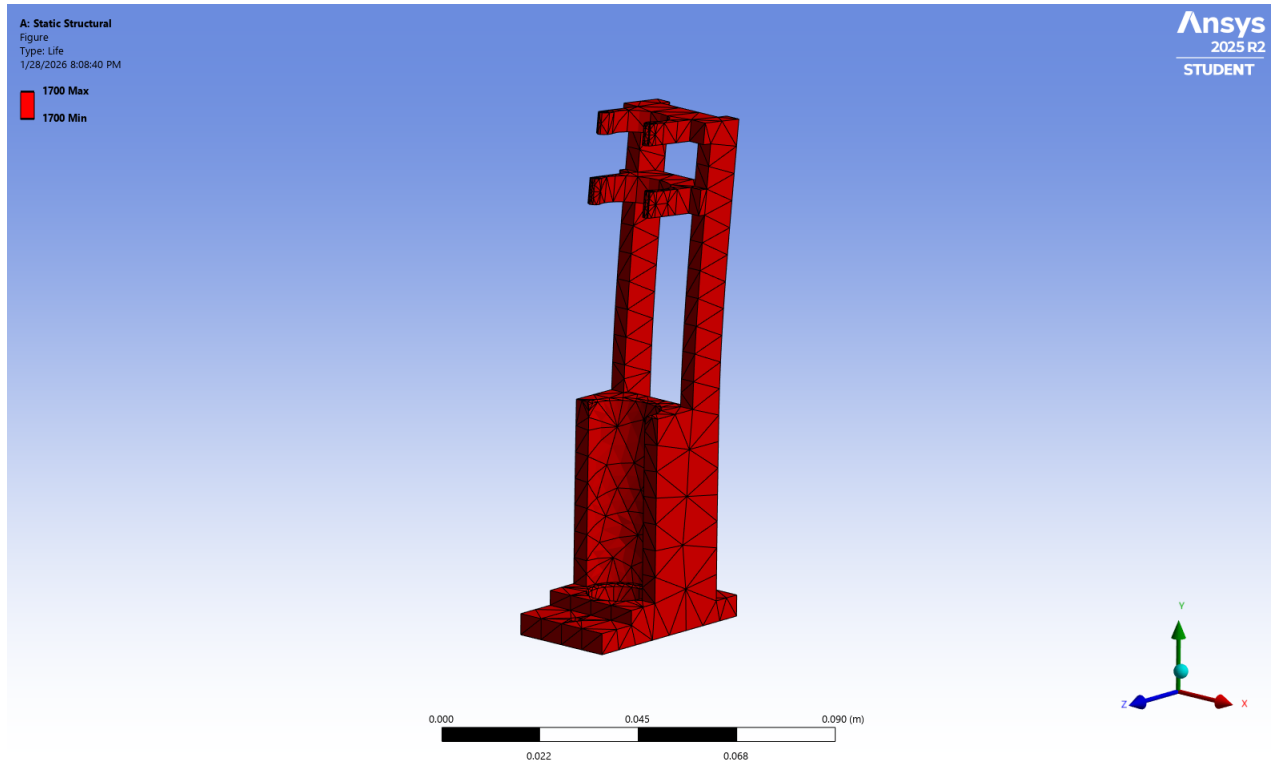


Figure 55: Life cycle of the slot made in PLA

The slot have the life cycle of 1700 uses, and while its lifetime is short, this is expected for a 3D-printed material like PLA, which also consistent with the tested maximum fatigue cycles of the same material is only exceeded a bit over 1700 cycles [2].

10 Conclusion

The Whiteboard Marker Ink Refiller machine has successfully demonstrated a cost-effective and sustainable solution for refilling markers. By integrating a pump-based lead screw mechanism for both the X and Z axes, the machine is able to automate the refilling process with high precision and efficiency. The following key outcomes were achieved:

- **Efficiency:** The system performs marker refilling within the desired time limit of 3 minutes, meeting the specified operational requirements.
- **Cost-Effectiveness:** The material cost for constructing the machine is well under 20, making it a viable alternative to disposable markers in educational institutions and offices.
- **Sustainability:** The design promotes sustainability by reducing plastic waste associated with disposable markers, offering a circular economy solution by allowing for repeated refills.
- **Mechanical Safety:** The reduction in mechanical friction and torque requirements, achieved by optimizing the offset distance in the Z-axis, improves the system's reliability and durability.

- User-Friendliness: The system is designed with simplicity in mind, using intuitive automation and safe components, making it easy for users to operate.

This project has proven to be an effective prototype for automated marker refilling , combining mechanical design, electronics, and sustainability. Further improvements can be made by incorporating customization ink choices , a user interface , and the full CNC-machining of parts to enhance precision and efficiency. The project serves as a promising step towards reducing waste and promoting sustainability in everyday products.

Future work could also include expanding the system to handle a greater variety of markers, as well as adding more advanced features like automatic ink-level detection and remote operation.

References

- [1] E. Tunca and H. Kafali, “Compression and three-point bending analyzes of aerospace sandwich composites produced with polymeric core materials with ansys,” *European Journal of Science and Technology*, 12 2021.
- [2] N. Ambhore, A. Dhumal, O. Kiran Borawake, V. Marathe, and D. Nalawade, “Fatigue analysis of polylactide manufactured by fused deposition method of 3d printing,” *Journal of Mines, Metals and Fuels*, pp. 957–964, 10 2024.
- [3] B. Rajakumar and V. Skm, “Evidence to support the mechanical advantage hypothesis of grasping at low force levels,” *Scientific Reports*, vol. 12, 12 2022.

A CNC Programming Report

Below is the G-code used for the milling operation:

```
1 %
2 O1000 (PLATE 1)
3 N100 (COMPENSATION-WEAR)
4 N102 (REV-0.70)
5 N104 (DEC-30-2025-11:04:49AM)
6
7
8 N106 (TOOL 10 - DIA 10.)
9 N108 (TOOL 5 - DIA 5.)
10 N110 (TOOL 6 - DIA 3.3)
11 N112 (TOOL 3 - DIA 3.)
12
13
14 N1 G90 G17 G40 G80 G00
15 N114 M06 T10 ()
16 N116 T5
17 N118 (P-contour)
18 N120 G00 G54 G90 X-35.5 Y5. S1500 M03
19 N122 G43 H10 Z120.
20 N124 S3500
21 N126 Z25.
22 N128 Z2.
23 N130 G01 Z-1.7 F300.
24 N132 X-95.5 F500.
25 N134 G03 X-95.5 Y-5. I0. J-5.
26 N136 G01 X24.5
27 N138 G03 X24.5 Y5. I0. J5.
28 N140 G01 X-35.5
29 N142 Z-3.4 F300.
30 N144 X-95.5 F500.
31 N146 G03 X-95.5 Y-5. I0. J-5.
32 N148 G01 X24.5
33 N150 G03 X24.5 Y5. I0. J5.
34 N152 G01 X-35.5
35 N154 Z-5.1 F300.
36 N156 X-95.5 F500.
37 N158 G03 X-95.5 Y-5. I0. J-5.
38 N160 G01 X24.5
39 N162 G03 X24.5 Y5. I0. J5.
40 N164 G01 X-35.5
41 N166 Z-6.8 F300.
42 N168 X-95.5 F500.
43 N170 G03 X-95.5 Y-5. I0. J-5.
44 N172 G01 X24.5
45 N174 G03 X24.5 Y5. I0. J5.
46 N176 G01 X-35.5
47 N178 Z-8.5 F300.
48 N180 X-95.5 F500.
49 N182 G03 X-95.5 Y-5. I0. J-5.
50 N184 G01 X24.5
51 N186 G03 X24.5 Y5. I0. J5.
52 N188 G01 X-35.5
53 N190 Z-10.2 F300.
54 N192 X-95.5 F500.
55 N194 G03 X-95.5 Y-5. I0. J-5.
56 N196 G01 X24.5
57 N198 G03 X24.5 Y5. I0. J5.
58 N200 G01 X-35.5
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59 N202 G00 Z25.
60 N204 M05
61 N206 M01
62
63 N2 G90 G17 G40 G80 G00
64 N208 M06 T5 ()
65 N210 T6
66 N212 (P-contour5)
67 N214 G00 G54 G90 X-158.407 Y-31.8728 S1500 M03
68 N216 G43 H5 Z120.
69 N218 S1500
70 N220 Z25.
71 N222 Z2.
72 N224 G01 X-158.9997 Y-34.25 Z1.7857 F300.
73 N226 X-158.407 Y-31.8728 Z1.5714
74 N228 X-158.9997 Y-34.25 Z1.3571
75 N230 X-158.407 Y-31.8728 Z1.1429
76 N232 X-158.9997 Y-34.25 Z0.9286
77 N234 X-158.407 Y-31.8728 Z0.7143
78 N236 X-158.9997 Y-34.25 Z0.5
79 N238 X-158.407 Y-31.8728 Z0.2857
80 N240 X-158.9997 Y-34.25 Z0.0714
81 N242 X-158.407 Y-31.8728 Z-0.1429
82 N244 X-158.9997 Y-34.25 Z-0.3571
83 N246 X-158.407 Y-31.8728 Z-0.5714
84 N248 X-158.9997 Y-34.25 Z-0.7857
85 N250 X-158.407 Y-31.8728 Z-1.
86 N252 X-158.9997 Y-34.25 F500.
87 N254 Y34.25
88 N256 X-141.75
89 N258 Y-34.25
90 N260 X-158.9997
91 N262 X-156.4997 Y-31.75
92 N264 Y31.75
93 N266 X-144.25
94 N268 Y-31.75
95 N270 X-156.4997
96 N272 X-153.9997 Y-29.25
97 N274 Y29.25
98 N276 X-146.75
99 N278 Y-29.25
100 N280 X-153.9997
101 N282 X-151.4997 Y-26.75
102 N284 Y26.75
103 N286 X-149.25
104 N288 Y-26.75
105 N290 X-151.4997
106 N292 G00 Z25.
107 N294 X-158.407 Y-31.8728
108 N296 Z1.
109 N298 G01 X-158.9997 Y-34.25 Z0.7857 F300.
110 N300 X-158.407 Y-31.8728 Z0.5714
111 N302 X-158.9997 Y-34.25 Z0.3571
112 N304 X-158.407 Y-31.8728 Z0.1429
113 N306 X-158.9997 Y-34.25 Z-0.0714
114 N308 X-158.407 Y-31.8728 Z-0.2857
115 N310 X-158.9997 Y-34.25 Z-0.5
116 N312 X-158.407 Y-31.8728 Z-0.7143
117 N314 X-158.9997 Y-34.25 Z-0.9286
118 N316 X-158.407 Y-31.8728 Z-1.1429
119 N318 X-158.9997 Y-34.25 Z-1.3571
120 N320 X-158.407 Y-31.8728 Z-1.5714

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124 N328 Y34.25
125 N330 X-141.75
126 N332 Y-34.25
127 N334 X-158.9997
128 N336 X-156.4997 Y-31.75
129 N338 Y31.75
130 N340 X-144.25
131 N342 Y-31.75
132 N344 X-156.4997
133 N346 X-153.9997 Y-29.25
134 N348 Y29.25
135 N350 X-146.75
136 N352 Y-29.25
137 N354 X-153.9997
138 N356 X-151.4997 Y-26.75
139 N358 Y26.75
140 N360 X-149.25
141 N362 Y-26.75
142 N364 X-151.4997
143 N366 G00 Z25.
144 N368 X81.25 Y0.
145 N370 Z2.
146 N372 G01 X80.5337 Y-2.3429 Z1.7857 F300.
147 N374 X81.25 Y0. Z1.5714
148 N376 X80.5337 Y-2.3429 Z1.3571
149 N378 X81.25 Y0. Z1.1429
150 N380 X80.5337 Y-2.3429 Z0.9286
151 N382 X81.25 Y0. Z0.7143
152 N384 X80.5337 Y-2.3429 Z0.5
153 N386 X81.25 Y0. Z0.2857
154 N388 X80.5337 Y-2.3429 Z0.0714
155 N390 X81.25 Y0. Z-0.1429
156 N392 X80.5337 Y-2.3429 Z-0.3571
157 N394 X81.25 Y0. Z-0.5714
158 N396 X80.5337 Y-2.3429 Z-0.7857
159 N398 X81.25 Y0. Z-1.
160 N400 X80.5337 Y-2.3429 F500.
161 N402 X81.2397 Y0.
162 N404 X81.25
163 N406 Y-34.25
164 N408 X70.75
165 N410 Y34.25
166 N412 X81.25
167 N414 Y0.
168 N416 X78.75
169 N418 Y-31.75
170 N420 X73.25
171 N422 Y31.75
172 N424 X78.75
173 N426 Y0.
174 N428 X76.25
175 N430 Y-29.25
176 N432 X75.75
177 N434 Y29.25
178 N436 X76.25
179 N438 Y0.
180 N440 G00 Z25.
181 N442 X81.25
182 N444 Z1.

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183 N446 G01 X80.5337 Y-2.3429 Z0.7857 F300.
184 N448 X81.25 Y0. Z0.5714
185 N450 X80.5337 Y-2.3429 Z0.3571
186 N452 X81.25 Y0. Z0.1429
187 N454 X80.5337 Y-2.3429 Z-0.0714
188 N456 X81.25 Y0. Z-0.2857
189 N458 X80.5337 Y-2.3429 Z-0.5
190 N460 X81.25 Y0. Z-0.7143
191 N462 X80.5337 Y-2.3429 Z-0.9286
192 N464 X81.25 Y0. Z-1.1429
193 N466 X80.5337 Y-2.3429 Z-1.3571
194 N468 X81.25 Y0. Z-1.5714
195 N470 X80.5337 Y-2.3429 Z-1.7857
196 N472 X81.25 Y0. Z-2.
197 N474 X80.5337 Y-2.3429 F500.
198 N476 X81.2397 Y0.
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200 N480 Y-34.25
201 N482 X70.75
202 N484 Y34.25
203 N486 X81.25
204 N488 Y0.
205 N490 X78.75
206 N492 Y-31.75
207 N494 X73.25
208 N496 Y31.75
209 N498 X78.75
210 N500 Y0.
211 N502 X76.25
212 N504 Y-29.25
213 N506 X75.75
214 N508 Y29.25
215 N510 X76.25
216 N512 Y0.
217 N514 G00 Z25.
218 N516 X158.9997 Y32.6
219 N518 Z2.
220 N520 G01 X157.7379 Y30.4999 Z1.7857 F300.
221 N522 X158.9997 Y32.6 Z1.5714
222 N524 X157.7379 Y30.4999 Z1.3571
223 N526 X158.9997 Y32.6 Z1.1429
224 N528 X157.7379 Y30.4999 Z0.9286
225 N530 X158.9997 Y32.6 Z0.7143
226 N532 X157.7379 Y30.4999 Z0.5
227 N534 X158.9997 Y32.6 Z0.2857
228 N536 X157.7379 Y30.4999 Z0.0714
229 N538 X158.9997 Y32.6 Z-0.1429
230 N540 X157.7379 Y30.4999 Z-0.3571
231 N542 X158.9997 Y32.6 Z-0.5714
232 N544 X157.7379 Y30.4999 Z-0.7857
233 N546 X158.9997 Y32.6 Z-1.
234 N548 X157.7379 Y30.4999 F500.
235 N550 X158.9895 Y32.5898
236 N552 X158.9997 Y32.6
237 N554 Y-32.6
238 N556 X119.75
239 N558 Y32.6
240 N560 X158.9997
241 N562 X156.4997 Y30.1
242 N564 Y-30.1
243 N566 X122.25
244 N568 Y30.1

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246 N572 X153.9997 Y27.6
247 N574 Y-27.6
248 N576 X124.75
249 N578 Y27.6
250 N580 X153.9997
251 N582 X151.4997 Y25.1
252 N584 Y-25.1
253 N586 X127.25
254 N588 Y25.1
255 N590 X151.4997
256 N592 X148.9997 Y22.6
257 N594 Y-22.6
258 N596 X129.75
259 N598 Y22.6
260 N600 X148.9997
261 N602 X146.4997 Y20.1
262 N604 Y-20.1
263 N606 X132.25
264 N608 Y20.1
265 N610 X146.4997
266 N612 X143.9997 Y17.6
267 N614 Y-17.6
268 N616 X134.75
269 N618 Y17.6
270 N620 X143.9997
271 N622 X141.4997 Y15.1
272 N624 Y-15.1
273 N626 X137.25
274 N628 Y15.1
275 N630 X141.4997
276 N632 G00 Z25.
277 N634 X158.9997 Y32.6
278 N636 Z1.
279 N638 G01 X157.7379 Y30.4999 Z0.7857 F300.
280 N640 X158.9997 Y32.6 Z0.5714
281 N642 X157.7379 Y30.4999 Z0.3571
282 N644 X158.9997 Y32.6 Z0.1429
283 N646 X157.7379 Y30.4999 Z-0.0714
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285 N650 X157.7379 Y30.4999 Z-0.5
286 N652 X158.9997 Y32.6 Z-0.7143
287 N654 X157.7379 Y30.4999 Z-0.9286
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291 N662 X157.7379 Y30.4999 Z-1.7857
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293 N666 X157.7379 Y30.4999 F500.
294 N668 X158.9895 Y32.5898
295 N670 X158.9997 Y32.6
296 N672 Y-32.6
297 N674 X119.75
298 N676 Y32.6
299 N678 X158.9997
300 N680 X156.4997 Y30.1
301 N682 Y-30.1
302 N684 X122.25
303 N686 Y30.1
304 N688 X156.4997
305 N690 X153.9997 Y27.6
306 N692 Y-27.6

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307 N694 X124.75
308 N696 Y27.6
309 N698 X153.9997
310 N700 X151.4997 Y25.1
311 N702 Y-25.1
312 N704 X127.25
313 N706 Y25.1
314 N708 X151.4997
315 N710 X148.9997 Y22.6
316 N712 Y-22.6
317 N714 X129.75
318 N716 Y22.6
319 N718 X148.9997
320 N720 X146.4997 Y20.1
321 N722 Y-20.1
322 N724 X132.25
323 N726 Y20.1
324 N728 X146.4997
325 N730 X143.9997 Y17.6
326 N732 Y-17.6
327 N734 X134.75
328 N736 Y17.6
329 N738 X143.9997
330 N740 X141.4997 Y15.1
331 N742 Y-15.1
332 N744 X137.25
333 N746 Y15.1
334 N748 X141.4997
335 N750 G00 Z25.
336 N752 M05
337 N754 M01
338
339 N3 G90 G17 G40 G80 G00
340 N756 M06 T6 ()
341 N758 T3
342 N760 (D-drill)
343 N762 G00 G54 G90 X-150.5 Y-30. S2000 M03
344 N764 G43 H6 Z120.
345 N766 S2000
346 N768 Z25.
347 N770 G98 G83 X-150.5 Y-30. Z-10. R2. Q0. F150.
348 N772 X-143.5
349 N774 Y30.
350 N776 X-150.5
351 N778 X72.5 Y-30.
352 N780 X79.5
353 N782 Y30.
354 N784 X72.5
355 N786 X129.5 Y-30.5
356 N788 Y-17.5
357 N790 Y17.5
358 N792 Y30.5
359 N794 X149.5
360 N796 Y17.5
361 N798 Y-17.5
362 N800 Y-30.5
363 N802 G80
364 N804 M05
365 N806 M01
366
367 N4 G90 G17 G40 G80 G00
368 N808 M06 T3 ()

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369 N810 T6
370 N812 (D-drill11)
371 N814 G00 G54 G90 X-139.25 Y36.75 S2500 M03
372 N816 G43 H3 Z120.
373 N818 S2500
374 N820 Z25.
375 N822 G98 G83 X-139.25 Y36.75 Z-2. R2. Q0. F150.
376 N824 Y-36.75
377 N826 X68.25
378 N828 Y36.75
379 N830 X83.75
380 N832 Y-36.75
381 N834 X117.25 Y-35.1
382 N836 Y35.1
383 N838 G80
384 N840 M05
385 N842 M01
386
387 N5 G90 G17 G40 G80 G00
388 N844 M06 T6 ()
389 N846 T3
390 N848 (D-drill12)
391 N850 G00 G54 G90 X-143. Y-44. S3000 M03
392 N852 G43 H6 Z120.
393 N854 S3000
394 N856 Z25.
395 N858 G98 G83 X-143. Y-44. Z-10. R2. Q0. F150.
396 N860 Y44.
397 N862 X72.
398 N864 Y-44.
399 N866 G80
400 N868 M05
401 N870 M01
402
403 N6 G90 G17 G40 G80 G00
404 N872 M06 T3 ()
405 N874 (P-contour2)
406 N876 G00 G54 G90 X-142.35 Y-44. S1500 M03
407 N878 G43 H3 Z120.
408 N880 S3500
409 N882 Z25.
410 N884 Z2.
411 N886 G01 Z-1. F150.
412 N888 G03 X-142.35 Y-44. I-0.65 J0. F300.
413 N890 G01 Z-2. F150.
414 N892 G03 X-142.35 Y-44. I-0.65 J0. F300.
415 N894 G01 Z-3. F150.
416 N896 G03 X-142.35 Y-44. I-0.65 J0. F300.
417 N898 G01 Z-4. F150.
418 N900 G03 X-142.35 Y-44. I-0.65 J0. F300.
419 N902 G01 Z-5. F150.
420 N904 G03 X-142.35 Y-44. I-0.65 J0. F300.
421 N906 G01 Z-6. F150.
422 N908 G03 X-142.35 Y-44. I-0.65 J0. F300.
423 N910 G01 Z-7. F150.
424 N912 G03 X-142.35 Y-44. I-0.65 J0. F300.
425 N914 G01 Z-8. F150.
426 N916 G03 X-142.35 Y-44. I-0.65 J0. F300.
427 N918 G01 Z-9. F150.
428 N920 G03 X-142.35 Y-44. I-0.65 J0. F300.
429 N922 G01 Z-10. F150.
430 N924 G03 X-142.35 Y-44. I-0.65 J0. F300.

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431 N926 G00 Z25.
432 N928 Y44.
433 N930 Z2.
434 N932 G01 Z-1. F150.
435 N934 G03 X-142.35 Y44. I-0.65 J0. F300.
436 N936 G01 Z-2. F150.
437 N938 G03 X-142.35 Y44. I-0.65 J0. F300.
438 N940 G01 Z-3. F150.
439 N942 G03 X-142.35 Y44. I-0.65 J0. F300.
440 N944 G01 Z-4. F150.
441 N946 G03 X-142.35 Y44. I-0.65 J0. F300.
442 N948 G01 Z-5. F150.
443 N950 G03 X-142.35 Y44. I-0.65 J0. F300.
444 N952 G01 Z-6. F150.
445 N954 G03 X-142.35 Y44. I-0.65 J0. F300.
446 N956 G01 Z-7. F150.
447 N958 G03 X-142.35 Y44. I-0.65 J0. F300.
448 N960 G01 Z-8. F150.
449 N962 G03 X-142.35 Y44. I-0.65 J0. F300.
450 N964 G01 Z-9. F150.
451 N966 G03 X-142.35 Y44. I-0.65 J0. F300.
452 N968 G01 Z-10. F150.
453 N970 G03 X-142.35 Y44. I-0.65 J0. F300.
454 N972 G00 Z25.
455 N974 X71.35
456 N976 Z2.
457 N978 G01 Z-1. F150.
458 N980 G03 X71.35 Y44. I0.65 J0. F300.
459 N982 G01 Z-2. F150.
460 N984 G03 X71.35 Y44. I0.65 J0. F300.
461 N986 G01 Z-3. F150.
462 N988 G03 X71.35 Y44. I0.65 J0. F300.
463 N990 G01 Z-4. F150.
464 N992 G03 X71.35 Y44. I0.65 J0. F300.
465 N994 G01 Z-5. F150.
466 N996 G03 X71.35 Y44. I0.65 J0. F300.
467 N998 G01 Z-6. F150.
468 N1000 G03 X71.35 Y44. I0.65 J0. F300.
469 N1002 G01 Z-7. F150.
470 N1004 G03 X71.35 Y44. I0.65 J0. F300.
471 N1006 G01 Z-8. F150.
472 N1008 G03 X71.35 Y44. I0.65 J0. F300.
473 N1010 G01 Z-9. F150.
474 N1012 G03 X71.35 Y44. I0.65 J0. F300.
475 N1014 G01 Z-10. F150.
476 N1016 G03 X71.35 Y44. I0.65 J0. F300.
477 N1018 G00 Z25.
478 N1020 Y-44.
479 N1022 Z2.
480 N1024 G01 Z-1. F150.
481 N1026 G03 X71.35 Y-44. I0.65 J0. F300.
482 N1028 G01 Z-2. F150.
483 N1030 G03 X71.35 Y-44. I0.65 J0. F300.
484 N1032 G01 Z-3. F150.
485 N1034 G03 X71.35 Y-44. I0.65 J0. F300.
486 N1036 G01 Z-4. F150.
487 N1038 G03 X71.35 Y-44. I0.65 J0. F300.
488 N1040 G01 Z-5. F150.
489 N1042 G03 X71.35 Y-44. I0.65 J0. F300.
490 N1044 G01 Z-6. F150.
491 N1046 G03 X71.35 Y-44. I0.65 J0. F300.
492 N1048 G01 Z-7. F150.

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493 N1050 G03 X71.35 Y-44. I0.65 J0. F300.
494 N1052 G01 Z-8. F150.
495 N1054 G03 X71.35 Y-44. I0.65 J0. F300.
496 N1056 G01 Z-9. F150.
497 N1058 G03 X71.35 Y-44. I0.65 J0. F300.
498 N1060 G01 Z-10. F150.
499 N1062 G03 X71.35 Y-44. I0.65 J0. F300.
500 N1064 G00 Z25.
501 N1066 (F-contour3)
502 N1068 S1500
503 N1070 X-158.3 Y43.7574
504 N1072 Z25.
505 N1074 Z2.
506 N1076 G01 Z-1.7 F300.
507 N1078 Y41.9574 F500.
508 N1080 G03 X-156.5 Y43.7574 I0. J1.8
509 N1082 G02 X-155.0355 Y47.2929 I5. J0.
510 N1084 G01 X-151.7929 Y50.5355
511 N1086 G02 X-148.2574 Y52. I3.5355 J-3.5355
512 N1088 G01 X-116.2082
513 N1090 G02 X-113.9721 Y51.4721 I0. J-5.
514 N1092 G01 X-95.239 Y42.1056
515 N1094 G03 X-94.7918 Y42. I0.4472 J0.8944
516 N1096 G01 X23.7918
517 N1098 G03 X24.239 Y42.1056 I0. J1.
518 N1100 G01 X42.9721 Y51.4721
519 N1102 G02 X45.2082 Y52. I2.2361 J-4.4721
520 N1104 G01 X88.2574
521 N1106 G02 X91.7929 Y50.5355 I0. J-5.
522 N1108 G01 X95.0355 Y47.2929
523 N1110 G03 X95.7426 Y47. I0.7071 J0.7071
524 N1112 G01 X151.5
525 N1114 G02 X156.5 Y42. I0. J-5.
526 N1116 G01 Y-42.
527 N1118 G02 X151.5 Y-47. I-5. J0.
528 N1120 G01 X95.7426
529 N1122 G03 X95.0355 Y-47.2929 I0. J-1.
530 N1124 G01 X91.7929 Y-50.5355
531 N1126 G02 X88.2574 Y-52. I-3.5355 J3.5355
532 N1128 G01 X45.2082
533 N1130 G02 X42.9721 Y-51.4721 I0. J5.
534 N1132 G01 X24.239 Y-42.1056
535 N1134 G03 X23.7918 Y-42. I-0.4472 J-0.8944
536 N1136 G01 X-94.7918
537 N1138 G03 X-95.239 Y-42.1056 I0. J-1.
538 N1140 G01 X-113.9721 Y-51.4721
539 N1142 G02 X-116.2082 Y-52. I-2.2361 J4.4721
540 N1144 G01 X-148.2574
541 N1146 G02 X-151.7929 Y-50.5355 I0. J5.
542 N1148 G01 X-155.0355 Y-47.2929
543 N1150 G02 X-156.5 Y-43.7574 I3.5355 J3.5355
544 N1152 G01 Y43.7574
545 N1154 G03 X-158.3 Y45.5574 I-1.8 J0.
546 N1156 G01 Y43.7574
547 N1158 Z-3.4 F300.
548 N1160 Y41.9574 F500.
549 N1162 G03 X-156.5 Y43.7574 I0. J1.8
550 N1164 G02 X-155.0355 Y47.2929 I5. J0.
551 N1166 G01 X-151.7929 Y50.5355
552 N1168 G02 X-148.2574 Y52. I3.5355 J-3.5355
553 N1170 G01 X-116.2082
554 N1172 G02 X-113.9721 Y51.4721 I0. J-5.

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555 N1174 G01 X-95.239 Y42.1056
556 N1176 G03 X-94.7918 Y42. I0.4472 J0.8944
557 N1178 G01 X23.7918
558 N1180 G03 X24.239 Y42.1056 I0. J1.
559 N1182 G01 X42.9721 Y51.4721
560 N1184 G02 X45.2082 Y52. I2.2361 J-4.4721
561 N1186 G01 X88.2574
562 N1188 G02 X91.7929 Y50.5355 I0. J-5.
563 N1190 G01 X95.0355 Y47.2929
564 N1192 G03 X95.7426 Y47. I0.7071 J0.7071
565 N1194 G01 X151.5
566 N1196 G02 X156.5 Y42. I0. J-5.
567 N1198 G01 Y-42.
568 N1200 G02 X151.5 Y-47. I-5. J0.
569 N1202 G01 X95.7426
570 N1204 G03 X95.0355 Y-47.2929 I0. J-1.
571 N1206 G01 X91.7929 Y-50.5355
572 N1208 G02 X88.2574 Y-52. I-3.5355 J3.5355
573 N1210 G01 X45.2082
574 N1212 G02 X42.9721 Y-51.4721 I0. J5.
575 N1214 G01 X24.239 Y-42.1056
576 N1216 G03 X23.7918 Y-42. I-0.4472 J-0.8944
577 N1218 G01 X-94.7918
578 N1220 G03 X-95.239 Y-42.1056 I0. J-1.
579 N1222 G01 X-113.9721 Y-51.4721
580 N1224 G02 X-116.2082 Y-52. I-2.2361 J4.4721
581 N1226 G01 X-148.2574
582 N1228 G02 X-151.7929 Y-50.5355 I0. J5.
583 N1230 G01 X-155.0355 Y-47.2929
584 N1232 G02 X-156.5 Y-43.7574 I3.5355 J3.5355
585 N1234 G01 Y43.7574
586 N1236 G03 X-158.3 Y45.5574 I-1.8 J0.
587 N1238 G01 Y43.7574
588 N1240 Z-5.1 F300.
589 N1242 Y41.9574 F500.
590 N1244 G03 X-156.5 Y43.7574 I0. J1.8
591 N1246 G02 X-155.0355 Y47.2929 I5. J0.
592 N1248 G01 X-151.7929 Y50.5355
593 N1250 G02 X-148.2574 Y52. I3.5355 J-3.5355
594 N1252 G01 X-116.2082
595 N1254 G02 X-113.9721 Y51.4721 I0. J-5.
596 N1256 G01 X-95.239 Y42.1056
597 N1258 G03 X-94.7918 Y42. I0.4472 J0.8944
598 N1260 G01 X23.7918
599 N1262 G03 X24.239 Y42.1056 I0. J1.
600 N1264 G01 X42.9721 Y51.4721
601 N1266 G02 X45.2082 Y52. I2.2361 J-4.4721
602 N1268 G01 X88.2574
603 N1270 G02 X91.7929 Y50.5355 I0. J-5.
604 N1272 G01 X95.0355 Y47.2929
605 N1274 G03 X95.7426 Y47. I0.7071 J0.7071
606 N1276 G01 X151.5
607 N1278 G02 X156.5 Y42. I0. J-5.
608 N1280 G01 Y-42.
609 N1282 G02 X151.5 Y-47. I-5. J0.
610 N1284 G01 X95.7426
611 N1286 G03 X95.0355 Y-47.2929 I0. J-1.
612 N1288 G01 X91.7929 Y-50.5355
613 N1290 G02 X88.2574 Y-52. I-3.5355 J3.5355
614 N1292 G01 X45.2082
615 N1294 G02 X42.9721 Y-51.4721 I0. J5.
616 N1296 G01 X24.239 Y-42.1056

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617 N1298 G03 X23.7918 Y-42. I-0.4472 J-0.8944
618 N1300 G01 X-94.7918
619 N1302 G03 X-95.239 Y-42.1056 I0. J-1.
620 N1304 G01 X-113.9721 Y-51.4721
621 N1306 G02 X-116.2082 Y-52. I-2.2361 J4.4721
622 N1308 G01 X-148.2574
623 N1310 G02 X-151.7929 Y-50.5355 I0. J5.
624 N1312 G01 X-155.0355 Y-47.2929
625 N1314 G02 X-156.5 Y-43.7574 I3.5355 J3.5355
626 N1316 G01 Y43.7574
627 N1318 G03 X-158.3 Y45.5574 I-1.8 J0.
628 N1320 G01 Y43.7574
629 N1322 Z-6.8 F300.
630 N1324 Y41.9574 F500.
631 N1326 G03 X-156.5 Y43.7574 I0. J1.8
632 N1328 G02 X-155.0355 Y47.2929 I5. J0.
633 N1330 G01 X-151.7929 Y50.5355
634 N1332 G02 X-148.2574 Y52. I3.5355 J-3.5355
635 N1334 G01 X-116.2082
636 N1336 G02 X-113.9721 Y51.4721 I0. J-5.
637 N1338 G01 X-95.239 Y42.1056
638 N1340 G03 X-94.7918 Y42. I0.4472 J0.8944
639 N1342 G01 X23.7918
640 N1344 G03 X24.239 Y42.1056 I0. J1.
641 N1346 G01 X42.9721 Y51.4721
642 N1348 G02 X45.2082 Y52. I2.2361 J-4.4721
643 N1350 G01 X88.2574
644 N1352 G02 X91.7929 Y50.5355 I0. J-5.
645 N1354 G01 X95.0355 Y47.2929
646 N1356 G03 X95.7426 Y47. I0.7071 J0.7071
647 N1358 G01 X151.5
648 N1360 G02 X156.5 Y42. I0. J-5.
649 N1362 G01 Y-42.
650 N1364 G02 X151.5 Y-47. I-5. J0.
651 N1366 G01 X95.7426
652 N1368 G03 X95.0355 Y-47.2929 I0. J-1.
653 N1370 G01 X91.7929 Y-50.5355
654 N1372 G02 X88.2574 Y-52. I-3.5355 J3.5355
655 N1374 G01 X45.2082
656 N1376 G02 X42.9721 Y-51.4721 I0. J5.
657 N1378 G01 X24.239 Y-42.1056
658 N1380 G03 X23.7918 Y-42. I-0.4472 J-0.8944
659 N1382 G01 X-94.7918
660 N1384 G03 X-95.239 Y-42.1056 I0. J-1.
661 N1386 G01 X-113.9721 Y-51.4721
662 N1388 G02 X-116.2082 Y-52. I-2.2361 J4.4721
663 N1390 G01 X-148.2574
664 N1392 G02 X-151.7929 Y-50.5355 I0. J5.
665 N1394 G01 X-155.0355 Y-47.2929
666 N1396 G02 X-156.5 Y-43.7574 I3.5355 J3.5355
667 N1398 G01 Y43.7574
668 N1400 G03 X-158.3 Y45.5574 I-1.8 J0.
669 N1402 G01 Y43.7574
670 N1404 Z-8.5 F300.
671 N1406 Y41.9574 F500.
672 N1408 G03 X-156.5 Y43.7574 I0. J1.8
673 N1410 G02 X-155.0355 Y47.2929 I5. J0.
674 N1412 G01 X-151.7929 Y50.5355
675 N1414 G02 X-148.2574 Y52. I3.5355 J-3.5355
676 N1416 G01 X-116.2082
677 N1418 G02 X-113.9721 Y51.4721 I0. J-5.
678 N1420 G01 X-95.239 Y42.1056

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679 N1422 G03 X-94.7918 Y42. I0.4472 J0.8944
680 N1424 G01 X23.7918
681 N1426 G03 X24.239 Y42.1056 I0. J1.
682 N1428 G01 X42.9721 Y51.4721
683 N1430 G02 X45.2082 Y52. I2.2361 J-4.4721
684 N1432 G01 X88.2574
685 N1434 G02 X91.7929 Y50.5355 I0. J-5.
686 N1436 G01 X95.0355 Y47.2929
687 N1438 G03 X95.7426 Y47. I0.7071 J0.7071
688 N1440 G01 X151.5
689 N1442 G02 X156.5 Y42. I0. J-5.
690 N1444 G01 Y-42.
691 N1446 G02 X151.5 Y-47. I-5. J0.
692 N1448 G01 X95.7426
693 N1450 G03 X95.0355 Y-47.2929 I0. J-1.
694 N1452 G01 X91.7929 Y-50.5355
695 N1454 G02 X88.2574 Y-52. I-3.5355 J3.5355
696 N1456 G01 X45.2082
697 N1458 G02 X42.9721 Y-51.4721 I0. J5.
698 N1460 G01 X24.239 Y-42.1056
699 N1462 G03 X23.7918 Y-42. I-0.4472 J-0.8944
700 N1464 G01 X-94.7918
701 N1466 G03 X-95.239 Y-42.1056 I0. J-1.
702 N1468 G01 X-113.9721 Y-51.4721
703 N1470 G02 X-116.2082 Y-52. I-2.2361 J4.4721
704 N1472 G01 X-148.2574
705 N1474 G02 X-151.7929 Y-50.5355 I0. J5.
706 N1476 G01 X-155.0355 Y-47.2929
707 N1478 G02 X-156.5 Y-43.7574 I3.5355 J3.5355
708 N1480 G01 Y43.7574
709 N1482 G03 X-158.3 Y45.5574 I-1.8 J0.
710 N1484 G01 Y43.7574
711 N1486 Z-10.2 F300.
712 N1488 Y41.9574 F500.
713 N1490 G03 X-156.5 Y43.7574 I0. J1.8
714 N1492 G02 X-155.0355 Y47.2929 I5. J0.
715 N1494 G01 X-151.7929 Y50.5355
716 N1496 G02 X-148.2574 Y52. I3.5355 J-3.5355
717 N1498 G01 X-116.2082
718 N1500 G02 X-113.9721 Y51.4721 I0. J-5.
719 N1502 G01 X-95.239 Y42.1056
720 N1504 G03 X-94.7918 Y42. I0.4472 J0.8944
721 N1506 G01 X23.7918
722 N1508 G03 X24.239 Y42.1056 I0. J1.
723 N1510 G01 X42.9721 Y51.4721
724 N1512 G02 X45.2082 Y52. I2.2361 J-4.4721
725 N1514 G01 X88.2574
726 N1516 G02 X91.7929 Y50.5355 I0. J-5.
727 N1518 G01 X95.0355 Y47.2929
728 N1520 G03 X95.7426 Y47. I0.7071 J0.7071
729 N1522 G01 X151.5
730 N1524 G02 X156.5 Y42. I0. J-5.
731 N1526 G01 Y-42.
732 N1528 G02 X151.5 Y-47. I-5. J0.
733 N1530 G01 X95.7426
734 N1532 G03 X95.0355 Y-47.2929 I0. J-1.
735 N1534 G01 X91.7929 Y-50.5355
736 N1536 G02 X88.2574 Y-52. I-3.5355 J3.5355
737 N1538 G01 X45.2082
738 N1540 G02 X42.9721 Y-51.4721 I0. J5.
739 N1542 G01 X24.239 Y-42.1056
740 N1544 G03 X23.7918 Y-42. I-0.4472 J-0.8944

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741 N1546 G01 X-94.7918
742 N1548 G03 X-95.239 Y-42.1056 I0. J-1.
743 N1550 G01 X-113.9721 Y-51.4721
744 N1552 G02 X-116.2082 Y-52. I-2.2361 J4.4721
745 N1554 G01 X-148.2574
746 N1556 G02 X-151.7929 Y-50.5355 I0. J5.
747 N1558 G01 X-155.0355 Y-47.2929
748 N1560 G02 X-156.5 Y-43.7574 I3.5355 J3.5355
749 N1562 G01 Y43.7574
750 N1564 G03 X-158.3 Y45.5574 I-1.8 J0.
751 N1566 G01 Y43.7574
752 N1568 S3500
753 N1570 Y41.4574 F300.
754 N1572 G03 X-156. Y43.7574 I0. J2.3
755 N1574 G02 X-154.682 Y46.9393 I4.5 J0.
756 N1576 G01 X-151.4393 Y50.182
757 N1578 G02 X-148.2574 Y51.5 I3.182 J-3.182
758 N1580 G01 X-116.2082
759 N1582 G02 X-114.1957 Y51.0249 I0. J-4.5
760 N1584 G01 X-95.4626 Y41.6584
761 N1586 G03 X-94.7918 Y41.5 I0.6708 J1.3416
762 N1588 G01 X23.7918
763 N1590 G03 X24.4626 Y41.6584 I0. J1.5
764 N1592 G01 X43.1957 Y51.0249
765 N1594 G02 X45.2082 Y51.5 I2.0125 J-4.0249
766 N1596 G01 X88.2574
767 N1598 G02 X91.4393 Y50.182 I0. J-4.5
768 N1600 G01 X94.682 Y46.9393
769 N1602 G03 X95.7426 Y46.5 I1.0607 J1.0607
770 N1604 G01 X151.5
771 N1606 G02 X156. Y42. I0. J-4.5
772 N1608 G01 Y-42.
773 N1610 G02 X151.5 Y-46.5 I-4.5 J0.
774 N1612 G01 X95.7426
775 N1614 G03 X94.682 Y-46.9393 I0. J-1.5
776 N1616 G01 X91.4393 Y-50.182
777 N1618 G02 X88.2574 Y-51.5 I-3.182 J3.182
778 N1620 G01 X45.2082
779 N1622 G02 X43.1957 Y-51.0249 I0. J4.5
780 N1624 G01 X24.4626 Y-41.6584
781 N1626 G03 X23.7918 Y-41.5 I-0.6708 J-1.3416
782 N1628 G01 X-94.7918
783 N1630 G03 X-95.4626 Y-41.6584 I0. J-1.5
784 N1632 G01 X-114.1957 Y-51.0249
785 N1634 G02 X-116.2082 Y-51.5 I-2.0125 J4.0249
786 N1636 G01 X-148.2574
787 N1638 G02 X-151.4393 Y-50.182 I0. J4.5
788 N1640 G01 X-154.682 Y-46.9393
789 N1642 G02 X-156. Y-43.7574 I3.182 J3.182
790 N1644 G01 Y43.7574
791 N1646 G03 X-158.3 Y46.0574 I-2.3 J0.
792 N1648 G01 Y43.7574
793 N1650 G00 Z25.
794 N1652 M05
795 N1654 G00 G28 G91 Z0
796 N1656 G00 G28 G91 X-15.0 Y0.
797 N1658 G90
798 N1660 M06 T10
799 N1662 M30
800 %

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Listing 1: Sample G-code for Square Pocket